04_Zomato

July 11, 2025

[350]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

[351]:
```
df1 = pd.read_csv('zomato.csv')
```

[352]:
```
df1.head() # collection of resturant
```

[352]:
```
                                                 url  \
0  https://www.zomato.com/bangalore/jalsa-banasha…
1  https://www.zomato.com/bangalore/spice-elephan…
2  https://www.zomato.com/SanchurroBangalore?cont…
3  https://www.zomato.com/bangalore/addhuri-udupi…
4  https://www.zomato.com/bangalore/grand-village…

                                             address                  name  \
0  942, 21st Main Road, 2nd Stage, Banashankari, …                 Jalsa
1  2nd Floor, 80 Feet Road, Near Big Bazaar, 6th …         Spice Elephant
2  1112, Next to KIMS Medical College, 17th Cross…        San Churro Cafe
3  1st Floor, Annakuteera, 3rd Stage, Banashankar…  Addhuri Udupi Bhojana
4  10, 3rd Floor, Lakshmi Associates, Gandhi Baza…          Grand Village

  online_order book_table   rate  votes                           phone  \
0          Yes        Yes  4.1/5    775    080 42297555\r\n+91 9743772233
1          Yes         No  4.1/5    787                      080 41714161
2          Yes         No  3.8/5    918                    +91 9663487993
3           No         No  3.7/5     88                    +91 9620009302
4           No         No  3.8/5    166  +91 8026612447\r\n+91 9901210005

       location            rest_type  \
0  Banashankari        Casual Dining
1  Banashankari        Casual Dining
2  Banashankari  Cafe, Casual Dining
3  Banashankari          Quick Bites
4  Basavanagudi        Casual Dining
```

```
                                          dish_liked  \
0  Pasta, Lunch Buffet, Masala Papad, Paneer Laja…
1  Momos, Lunch Buffet, Chocolate Nirvana, Thai G…
2  Churros, Cannelloni, Minestrone Soup, Hot Choc…
3                                       Masala Dosa
4                              Panipuri, Gol Gappe

                        cuisines approx_cost(for two people)  \
0  North Indian, Mughlai, Chinese                         800
1     Chinese, North Indian, Thai                         800
2          Cafe, Mexican, Italian                         800
3     South Indian, North Indian                          300
4       North Indian, Rajasthani                          600

                                      reviews_list menu_item  \
0  [('Rated 4.0', 'RATED\n  A beautiful place to …        []
1  [('Rated 4.0', 'RATED\n  Had been here for din…        []
2  [('Rated 3.0', "RATED\n  Ambience is not that …        []
3  [('Rated 4.0', "RATED\n  Great food and proper…        []
4  [('Rated 4.0', 'RATED\n  Very good restaurant …        []

  listed_in(type) listed_in(city)
0          Buffet    Banashankari
1          Buffet    Banashankari
2          Buffet    Banashankari
3          Buffet    Banashankari
4          Buffet    Banashankari
```

[353]:
```
df1['menu_item'].value_counts() # observation: there are many as (39617) empty
```

[353]:
```
[]
39617
['Butter Chicken Pizza', 'Bombay Veggie Burger', 'Tawa Paneer Burger', 'Oh So
Cheesy Burger', 'Barbecue Chicken Burger', 'Peri Peri Chicken Burger', 'BBQ
Lamb', 'Cheesy Chicken', 'Crunchy Ferrero', 'Lots of Nuts', 'Kadhai Paneer
Pizza', 'Chilli Paneer and Mushroom Pizza', 'Chipotle Veggie Pizza', 'Creamy
Cheese Pizza', 'Southern Veggie Korma Pizza', 'Popeye, The Corny Man Pizza',
'Paneer Tikka Pizza', 'Desi Margherita Pizza', 'Roasted Paneer with Mustard
Pizza', 'Bangalore Express Pizza', 'Italian Chaska Pizza', 'Mayo and Cheese
Pizza', 'Classic Margherita', 'Cajun Hawaiian Pizza', 'Butter Chicken Pizza',
'Lasooni Bhuna Murg Pizza', 'Chilly Chicken Pizza', 'Hot Garlic Chicken Pizza',
'Murg Barbecue Pizza', 'Mediterranean Mutton Keema Pizza', 'Super Green Burger',
'Quinoa Black Bean Burger', 'Bombay Veggie Burger', 'Tawa Paneer Burger', 'Oh So
Cheesy Burger', 'Barbecue Chicken Burger', 'Chipotle Lamb Burger', 'Crumb Fried
Fish Burger', 'Peri Peri Chicken Burger', 'BBQ Lamb', 'Cheesy Chicken', 'Picnic
Chicken Burger', 'Korean Grilled Chicken with Kimchi', 'Kiddy Kat Strawberry',
'Kiddy Kat Chocolate', 'Bubblegum', 'Oreo Shake', 'Caramel Shake', 'Chocolate
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Shake', 'Strawberry Shake', 'Vanilla Shake', 'Cold Coffee', 'Virgin Mojito', 'Iced Tea', 'Hot Chocolate', 'Cappuccino', 'Latte', 'Espresso Shot', 'Brownie Shake', 'Red Velvet Cheesecake Shake', 'Oreo Cheesecake Shake', 'Salted Caramel Popcorn', 'Coconut Crumble', 'Coffee Crunch', 'Bira', 'Thanda Paan', 'Sticky Toffee Pudding', 'Blueberry Cheesecake', 'Nutella Cheesecake', 'Caramel Custard with Strawberry or Figs', 'Red Velvet Cheesecake', 'Tiramisu', 'Zesty Vanilla and Tipsy Brownie', 'Oreo Explosion', 'Chocolate Almond Crumble', 'Crunchy Ferrero', "Stoner's Chocolate Decadence", 'Brownie Bash', 'Chocolate Lava', 'Dark Chocolate Caramel', 'Chocolate Overload', 'Lots of Nuts', 'Peanut Butter Crunch', 'Nutty Fudgy', 'Lemon Pistachio Biscotti', 'Frutilicious Ice Cream', 'Strawberry Tease Ice Cream', 'Mango Tango Ice Cream', 'Lychee Lovers Ice Cream', 'Monkey Business Ice Cream', 'Apple Crumble Ice Cream', 'Blueberry Bliss Ice Cream']
11
['Avil Milk', 'Oreo Shake', 'Chocolate Shake', 'Royal Falooda', 'Fruits Salad with Ice Cream', 'Fruits Salad with Ice Cream', 'Blackcurrant Ice Cream', 'Spanish Delight Ice Cream', 'Chillout Special Ice Cream', 'Chocolate Ice Cream', 'Butterscotch Ice Cream', 'Pista Ice Cream', 'Strawberry Ice Cream', 'Vanilla Ice Cream', 'Royal Falooda', 'Strawberry Falooda', 'Chillout Special Falooda', 'Lemon Juice', 'Mint Lemon Juice', 'Ginger Lemon Juice', 'Lemon Soda', 'Mint Lemon Soda', 'CAP Juice', 'Kiwi Juice', 'Carrot Juice', 'Pomegranate Juice', 'Watermelon Juice', 'Muskmelon Juice', 'Anjeer Juice', 'Pure Strawberry Juice', 'Papaya Juice', 'Orange Juice', 'Mosambi Juice', 'Grape Juice', 'Apple Juice', 'Pineapple Juice', 'Banana Juice', 'Rose Milk', 'Chikoo Shake', 'Sharja Shake', 'Kiwi Shake', 'Oreo Shake', 'Raspberry Shake', 'Pomegranate Shake', 'Mango Shake', 'Muskmelon Shake', 'Papaya Shake', 'Apple Shake', 'Berries Shake', 'Spanish Delight Shake', 'Blackcurrant Shake', 'Chillout Special Shake', 'Chocolate Shake', 'Butterscotch Shake', 'Strawberry Shake', 'Vanilla Shake', 'Pista Shake', 'Cold Coffee Shake', 'Avil Milk', 'Special Milk', 'Kulki Sarbath', 'Sweet Lassi', 'Chocolate Lassi', 'Mango Lassi', 'Banana Lassi', 'Strawberry Lassi']
10
['Chicken Cheese Burger', 'Chicken Billys BIg Burger', 'French Fries', 'Alfredo Veg Pasta', 'Brownie Waffles', 'Veg Burger', 'Veg Cheese Burger', 'Veg Caramelized onion Burger', 'Mushroom Classic Burger', 'Veg English Cheddar Cheese Burger', 'Veg Chipotle Corn Burger', 'Veg Billys BIg Burger', 'Chicken Burger', 'Chicken Cheese Burger', 'Chicken Caramelized Onion Burger', 'Chicken Mushroom Burger', 'Chicken English Cheddar Cheese Burger', 'Chicken Chipotle Corn Burger', 'Chicken Billys BIg Burger', 'Chicken Nuggets Burger', 'Veg Grilled Sandwich', 'Paneer Burji Sandwich', 'Toasted Paneer Sandwich', 'Grilled Mushroom Sandwich', 'Corn and Cheese Sandwich', 'Egg Sandwich', 'Egg Chutney Sandwich', 'Chicken Mayo Sandwich', 'Chicken Cheese Sandwich', 'Hotdog Sandwich', 'Tuna Sandwich', 'Tangy Tomato Pasta', 'Spicy Penne Pasta', 'Alfredo Veg Pasta', 'Smoked Chicken Pasta', 'Alfredo Chicken Pasta', 'French Fries', 'Veg', 'Chicken', 'Fish Finger', 'Vegetable Omelette', 'Masala Omelette', 'Spanish Omelette', 'Cheese Omelette', 'Paneer Omelette', 'Ultimate Vegan Shakes', 'Bittersweet Symphony Shakes', 'The White Paradise Shakes', 'Deep Dark

Abyss Shakes', 'Irene Adlers Mystery Shakes', 'Red Weather Parody Shakes', 'Mind palace Shakes', 'Tower of Baskerville Shakes', 'Nights Demon Shakes', 'Black and White Romance Shakes', 'Day Dream Shakes', 'Sherlocks Addiction Shakes', 'Mrs Hudsons Secret Shakes', 'Classic Waffles', 'Caramel Waffles', 'Choco Treat Waffles', 'Brownie Waffles', 'Nutella Crunch Waffles', 'Snowflakes Waffles', 'Rocky Road Waffles', 'Lights Out Waffles']
9
['Students Veg Combo', 'Office Veg Combo', 'Students Non Veg Combo', 'Office Non Veg Combo', 'Mughlai Non Veg Combo', 'Aloo Paratha Combo', 'Poori Aloo Jeera Combo', 'Poori Aloo Tomato Combo', 'Paneer Paratha Combo', 'Students Veg Combo', 'Office Veg Combo', 'Students Non Veg Combo', 'Office Non Veg Combo', 'Mughlai Non Veg Combo', 'Shami Kebab Roll', 'Mughlai Biryani Combo [Veg]', 'Nawabi Thali Combo', 'The Lajawab Gravy Combo [Veg]', 'Mughlai Meal For Two [Veg]', 'The After Party Starters', 'Royal Feast For Four [Veg]', 'The Galawati Roll', 'Mughlai Biryani Combo [Non Veg]', 'The Nihari Combo [Chicken]', 'The Nihari Combo (Mutton)', 'The Lajawab Gravy Combo [Non Veg]', 'Mughlai Meal For Two [Non-Veg]', 'Biryani Blast', 'Royal Feast For Four [Non-Veg]', 'Baby Corn 65', 'Chilli Baby Corn', 'Mushroom 65', 'Chilli Mushroom', 'Veg Seekh Kebab', 'Kasturi Paneer Tikka', 'Paneer 65', 'Paneer Angara', 'Chilli Paneer', 'Hara Bhara Kebab', 'Paneer Tikka', 'Paneer Malai Tikka', 'Tandoori Veg Platter', 'Tandoori Chicken', 'Mutton Shami Kebab [Signature]', 'Mutton Galouti Kebab [Signature]', 'Chicken 65', 'Chicken Seekh Kebab', 'Chicken Tikka', 'Chilli Chicken', 'Chicken Angara', 'Chicken Kasturi Kebab', 'Chicken Malai Tikka', 'Mutton Sultani Seekh Kebab', 'Non Veg Tandoori Platter', 'Aloo Tomato Gravy', 'Aloo Capsicum Dry', 'Aloo Jeera Dry', 'Dal Fry', 'Dal Tadka', 'Rajma', 'Chole', 'Dal Makhani [Signature]', 'Lipta Mushroom Masala', 'Veg Kolhapuri', 'Chole Curry', 'Paneer Bhurji', 'Miloni Tarkari [Signature]', 'Mixed Veg', 'Palak Paneer', 'Diwani Handi', 'Mattar Paneer', 'Paneer Do Pyaza', 'Butter Paneer', 'Kadai Paneer [Signature]', 'Paneer Tikka Masala [Signature]', 'Methi Malai Paneer', 'Paneer Lababdar [Signature]', 'Murg Musallam', 'Egg Bhurji', 'Egg Curry', 'Chicken Nihari [Signature]', 'Chicken Curry', 'Chicken Do Pyaza', 'Chicken Methi Khas', 'Hyderabadi Chicken', 'Chicken Hara Pyaza', 'Chicken Kali Mirch', 'Chicken Kohlapuri', 'Kadai Chicken', 'Chicken Lababdar', 'Chicken Lajawab [Signature]', 'Chicken Tikka Masala', 'Chicken Boti Kebab Masala', 'Butter Chicken', 'Mughlai Chicken Masala [Signature]', 'Mutton Nihari [Signature]', 'Mutton Do Pyaza', 'Mutton Korma', 'Mutton Rogan Josh', 'Mughlai Mutton Masala', 'Mutton Boti Kebab Masala [Signature]', 'Murg Musallam', 'Steamed Rice', 'Jeera Rice', 'Ghee Rice', 'Khushka', 'Pulao', 'Veg Biryani', 'Mughlai Chicken Dum Biryani [Signature]', 'Chicken Tikka Biryani', 'Mughlai Mutton Dum Biryani [Signature]', 'Tandoori Roti', 'Lachha Paratha', 'Naan', 'Amritsari Kulcha', 'Garlic Naan', 'Missi Roti', 'Mughlai Paratha', 'Poori', 'Mughlai Bread Basket', 'Papad', 'Curd', 'Boondi Raita', 'Mixed Raita', 'Masala Papad', 'Pineapple Raita', 'Paneer Roll', 'Hawker Style Boiled Egg', 'Masala Omelette', 'Chicken Roll', 'Chicken Boti Roll', 'Shami Roll', 'The Galouti Roll [Signature]', 'Mutton Boti Roll', 'Gulab Jamun [2 Pieces]', 'Ice Cream Scoop [single]', 'Rice Kheer', 'Awadhi Kheer [Signature]', 'Gajar Ka Halwa', 'Lachhedar Rabri', 'Shahi Tukda [Signature]', 'Thums Up [750 Ml]', 'Mineral

Water [1 Litre]', 'Buttermilk', 'Lime Water', 'Thums Up (600 Ml)', 'Jal Jeera', 'Lassi', 'Lime Water Salt', 'Lime Water Sweet', 'Chocolate Lassi', 'Strawberry Lassi', 'Alphonso Mango Lassi', 'Thandai', 'Mango Lassi'] 9

…

['Veg Lasagna', 'Peri Peri Paneer', 'Chicken Steak', 'Schezwan Veg Noodles', 'Chicken Chilly Coriander', 'Plain French Fries', 'Crispy Corn', 'Nachos', 'Chicken Wings', 'Non Veg Alfredo Pasta', 'Farmhouse Pizza', 'Tomato Basil Soup', 'Veg Minestrone Soup', 'Cream of Mushroom Soup', 'Chicken Minestrone Soup', 'Cream of Chicken Soup', 'Veg Greek Salad', 'Veg Ceasar Salad', 'Veg Mexican Salad', 'Veg Russian Salad', 'Chicken Greek Salad', 'Chicken Ceasar Salad', 'Chicken Mexican Salad', 'Chicken Russian Salad', 'Gobi Manchurian', 'Gobi Chilly', 'Mushroom Chilly', 'Paneer 65', 'Paneer Chilly', 'Honey Chilly Potato', 'Egg Chilly', 'Egg Manchurian', 'Chicken Chilly Coriander', 'Fish Chilly Coriander', 'Chilly Fish', 'Plain French Fries', 'Masala French Fries', 'Crispy Corn', 'Bruschetta', 'Plain Garlic Bread', 'Cheese Garlic Bread', 'Pizza Sticks', 'Nachos', 'Chicken Wings', 'Fish Fingers', 'Veg Thai Green Curry', 'Veg Thai Red Curry', 'Veg Lasagna', 'Peri Peri Paneer', 'Mexican Cottage Chesse', 'Chicken Thai Green Curry', 'Chicken Thai Red Curry', 'Chicken Lasagna', 'Peri Peri Chicken', 'Chicken Steak', 'Grilled Fish in Lemon Butter Sauce', 'Chicken Stroganoff', 'Grilled Fish in Lemon Butter Sauce', 'Fish and Chips', 'Veg Alfredo Pasta', 'Veg Arrabiata Pasta', 'Veg Burnt Garlic Pasta', 'Veg Spaghetti Agli-e-Olio', 'Veg Basil Pesto Pasta', 'Non Veg Basil Pesto Pasta', 'Non Veg Spaghetti Agli-e-Olio', 'Non Veg Alfredo Pasta', 'Non Veg Arrabiata Pasta', 'Non Veg Burnt Garlic Pasta', 'Margherita Pizza', 'Farmhouse Pizza', 'Veg Basil Pesto Pizza', "Chef's Inhouse Paneer Pizza", "Chef's Inhouse Chicken Pizza", 'Non Veg Basil Pesto Pizza', 'Chicken Tikka Pizza', 'Veg Fried Rice', 'Schezwan Fried Rice', 'American Fried Rice', 'Chicken Fried Rice', 'Schezwan Chicken Fried Rice', 'American Chicken Fried Rice', 'Veg Noodles', 'American Noodles Veg', 'Schezwan Veg Noodles', 'Chicken Noodles', 'Schezwan Chicken Noodles', 'American Chicken Noodles', 'Veg Burger', 'Paneer Burger', 'Chicken Burger', 'Peri Peri Chicken Burger', 'Mixed Veg Wrap', 'Paneer Tikka Wrap', 'Egg and Cheese Wrap', 'Chicken Tikka Wrap']

1

['Konaseema Kodi Vepudu', 'Boneless Kodi Chips', 'Bangla Kodi', 'Kodi Roast', 'Jeedipappu Kodi Pakodi', 'Bamboo Chicken', 'Pachmirchi Kodi Kabab', 'Godavari Royyala Vepudu', 'Pappu Charu Annam', 'Gutti Vankaya Pulao', 'Ulavacharu Veg Pulao', 'Daddojanam', 'Raju Gari Kodi Pulao', 'Kodi Vepudu Pulao', 'Rayala Vari Kodi Pulao', 'Avakai Kodi Pulao', 'Boneless Pachi Mirchi Kodi Pulao', 'Gadwal Kodi Pulao', 'Royyala Vepudu Pulao', 'Chicken Dum Biryani', 'Mutton Dum Biryani', 'Ulavacharu', 'Gongura Kodi Kura', 'Veg Hot and Sour Soup', 'Veg Sweet Corn Soup', 'Veg Manchow Soup', 'Veg Lemon Coriander Soup', 'Non Veg Hot and Sour Soup', 'Non Veg Sweet Corn Soup', 'Non Veg Manchow Soup', 'Non Veg Lemon Coriander Soup', 'Garden Green Salad', 'Curd Onion Salad', 'Veg Seek Kabab', 'Hara Bhara Kabab', 'Paneer Tikka', 'Mokkajonna Vepudu', 'Veg Pandu Mirapakai', 'Chilly Paneer', 'Veg Crispy', 'Chilly Mushroom', 'Corn Kernel', 'Konaseema Kodi Vepudu', 'Boneless Kodi Chips', 'Bangla Kodi', 'Boneless Kotha Kodi', 'Kodi Roast', 'Jeedipappu Kodi Pakodi', 'Kodi Keema Balls', 'Bamboo Chicken',

'Boneless Peanut Chicken', 'Boneless Chinese Chilly Chicken', 'Boneless Schezwan Chicken', 'Boneless Crispy Chilly Chicken', 'Gogura Kodi Kabab', 'Pachmirchi Kodi Kabab', 'Pandu Mirapakaya Kodi Kabab', 'Murgh Malai Kabab', 'Boneless Mamsam Keema Balls', 'Mamsam Roasted', 'Boneless Konaseema Mamsam Vepudu', 'Boneless Mamsam Kaju Vepudu', 'Boneless Mamsam Miryala Vepudu', 'Boneless Korameenu Vepudu', 'Godavari Royyala Vepudu', 'Boneless Apollo Fish', 'Boneless Chilly Fish', 'Boneless Kolleti Chapala Vepudu', 'Loose Prawns', 'Chilly Prawns', 'Dal Fry', 'Dal Tadka', 'Veg Kandhari', 'Mixed Veg Curry', 'Methi Chaman', 'Paneer Butter Masala', 'Paneer Tikka Masala', 'Palak Paneer', 'Achari Paneer', 'Baby Corn Kaju Masala', 'Mushroom Tomato Curry', 'Chicken Masala', 'Boneless Chicken Chatpata', 'Boneless Butter Chicken', 'Punjabi Chicken', 'Boneless Methi Chicken', 'Ulavacharu', 'Lamb Rogan Josh', 'Andhra Kodi Kura', 'Miriyala Kodi Kura', 'Avakai Kodi Kura', 'Gongura Kodi Kura', 'Telangana Kodi Kura', 'Mamsam Pappucharu', 'Telangana Mamsam Kura', 'Keema Kura', 'Gongura Mamsam', 'Royyala Iguru', 'Gongura Royyalu', 'Phulka', 'Tandoori Roti', 'Naan', 'Butter Naan', 'Garlic Naan', 'Plain Paratha', 'Stuffed Paratha', 'Masala Kulcha', 'Stuffed Egg Paratha', 'Stuffed Keema Paratha', 'Pappu Charu Annam', 'Gutti Vankaya Pulao', 'Avakai Veg Pulao', 'Paneer Tikka Pulao', 'Pachi Mirchi Veg Pulao', 'Ulavacharu Veg Pulao', 'Daddojanam', 'Curd Rice', 'Plain Rice', 'Egg White Pulao', 'Raju Gari Kodi Pulao', 'Kodi Vepudu Pulao', 'Rayala Vari Kodi Pulao', 'Avakai Kodi Pulao', 'Boneless Pachi Mirchi Kodi Pulao', 'Gadwal Kodi Pulao', 'Pachi MirchiKeema Pulao', 'Mamsam Vepudu Pulao', 'Keema Vepudu Pulao', 'Pachi Mirchi Royyala Pulao', 'Royyala Vepudu Pulao', 'Veg Biryani', 'Chicken Dum Biryani', 'Mutton Dum Biryani', 'Veg Fried Rice', 'Veg Schezwan Fried Rice', 'Paneer Mashed Rice', 'Egg Fried Rice', 'Chicken Fried Rice', 'Schezwan Chicken Fried Rice', 'Mixed Fried Rice', 'Veg Soft Noodles', 'Veg Schezwan Noodles', 'Chicken Noodles', 'Schezwan Chicken Noodles', 'Omelette', 'Curd', 'Apricot Delight', 'Buttermilk', 'No.1 Buttermilk', 'Mineral Water', 'Fresh Lime Water', 'Fresh Lime Soda', 'Lassi'] 1
['ST. Johns Chicken Parcel', 'BBQ Chicken', 'Ratatouille', 'Humptom Steak', 'Steak Diane', 'Oppure Steak', 'Beef Stroganoff', 'Fried Ice Cream', 'Caramel Custard', 'Peri Peri Fries', 'Ultimate Veg Nachos', 'Spicy Moroccan Egg [6 Pieces]', 'Ultimate Chicken Nachos', 'Steak Burger', 'Beef Burger', 'Chicken Burger', 'Arrabbiata Pasta', 'Alfredo Pasta', 'Mushroom Cappuccino Soup', 'Minesstrone Soup', 'Cock A Leekie Soup', 'Lamb Hungarian Soup', 'Peperonata Salad', 'Mediterranean Pasta Salad', 'Classic Caesar Salad', 'Roasted Beef Salad [York ST. Special]', 'York Special Tuna Salad', 'Chefs Special Salad', 'Ultimate Veg Nachos', 'Penang Crispy Paneer', 'Two In Punch [6 Pieces]', 'Crispy Onion Rings', 'Spicy Moroccan Egg [6 Pieces]', 'Ultimate Chicken Nachos', 'Chicken Fingers [6 Pieces]', 'American Corn Cheese Bolls [6 Pieces]', 'Grilled Harissa Prawns', 'Cajun Spiced Fish Fingers [8 Pieces]', 'Prawns Olive Skewers [3 Skewers]', 'Panko Crusted Chicken [6 Pieces]', 'Deep Fried Prawns [8 Pieces]', "Fisherman's Choice", 'Vegetable Shashuk Forage', 'Vegetable Moussaka', 'Ratatouille', 'Boston Bake', 'Vegetable Steak', 'Vegetable Stir Fry', 'Mexican Green Rice with Tortilla Chips and Salsa', 'Stuffed Capsicum and Tomato Morray', 'Humptom Steak', 'Steak Diane', 'Oppure Steak', 'Beef Stroganoff', 'Arrabbiata Pasta', 'Alfredo Pasta', 'Basilica Pasta', 'Paprika Pasta', 'Marinara Pasta',

'Bolognasie Sauce', 'Veg Burger', 'Roasted Vegetable Burger', 'Steak Burger', 'Beef Burger', 'Chicken Burger', 'Lamb Burger', 'Roasted Vegetable Sandwich', 'Chicken Grilled Sandwich', 'BBQ Chicken Sandwich', 'Beef Grilled Sandwich', 'Tuna Grilled Sandwich', 'Garlic Toast', 'Fries', 'Peri Peri Fries', 'Cheese Fries', 'Cheese Toast', 'Chilli Cheese Toast', 'Wiener Schnitzel', 'ST. Johns Chicken Parcel', 'Triple Decker Chicken', 'Black Pepper Chicken', 'BBQ Chicken', 'Oven Roasted Herb Chicken', 'Lemon Basil Chicken Roller', 'Herbs Crusted Grilled Fish', 'Fish N Chips with Peas', "Fisherman's Bake", 'Grilled Fish with Lemon Butter Sauce', 'Fried Ice Cream', 'Caramel Custard', 'Proline Suffle', 'Panna Cotta', 'Sizzling Brownie', 'Green Apple Mojito', 'Orange Mojito', 'Blue Curacao Mojito', 'Strawberry Mojito', 'Passion Fruits Mojito', 'Peach Iced Tea', 'Lemon Iced Tea', 'Espresso Single Coffee', 'Espresso Double Coffee', 'Americano Coffee', 'Cappuccino Coffee', 'Cafe Latte Coffee', 'Cafe Mocha Coffee', 'Hot Chocolate Coffee', 'Iced Latte Cold Coffee', 'Mocha Iceberg Cold Coffee', 'Frozen Milano Cold Coffee', 'Brownie Shake Cold Coffee', 'Ferrero Rocher Shake Cold Coffee', 'Frozen Mango Frappe', 'Frozen Strawberry Frappe', 'LItchi Frappe', 'Kiwi Frappe', 'Mango Kiwi Smoothie', 'Ultimate Berry Shake', 'Basil Spitzer']
1
['Crispy Paneer Burger', 'Veg Burger', 'Egg Burger', 'Chicken Burger', 'Crispy Chicken Burger', 'Mutton Burger', 'Orange Juice', 'Oreo Milkshake', 'Chicken Burger Combo', 'Talayani Subwich', 'Milano Subwich', 'Rambo Subwich', 'Secret Garden Subwich', 'Talayani Subwich', 'Milano Subwich', 'Rambo Subwich', 'Veg Burger Combo', 'Crispy Paneer Burger Combo', 'Chicken Burger Combo', 'Mutton Burger Combo', 'Crispy Paneer Burger', 'Veg Burger', 'Egg Burger', 'Chicken Burger', 'Crispy Chicken Burger', 'Mutton Burger', 'Plain Omelette', 'Egg White Omelette', 'Masala Omelette', 'Chicken Omelette', 'Chicken Ham Omelette', 'Vanilla Ice Cream', 'Butterscotch Ice Cream', 'Strawberry Ice Cream', 'Mango Ice Cream', 'Chocolate Ice Cream', 'Vanilla and Butterscotch Sundae', 'Dry fruit Dark chocolate Ice Cream', 'Strawberry and Mango Delight', 'Orange Juice', 'Mosambi Juice', 'Watermelon Juice', 'Muskmelon Juice', 'Beetroot Juice', 'Pineapple Juice', 'Grape Juice', 'Fresh Lime', 'Fresh Lime Soda', 'Mango Milkshake', 'Pineapple Milkshake', 'Strawberry Milkshake', 'Vanilla Milkshake', 'Butterscotch Milkshake', 'Cold Chocolate Milkshake', 'Cold Coffee Milkshake', 'Oreo Milkshake', 'Pomegranate Milkshake', 'Orange Pure Juice', 'Mosambi Pure Juice', 'Pomegranate Pure Juice', 'Pineapple Pure Juice', 'Grape Pure Juice', 'Watermelon Pure Juice']
1
['Veg Mudde Combo', 'Veg Mini Meal Combo', 'Veg Nammura Oota Combo', 'Non Veg Mudde Oota Combo', 'Chicken Biryani Combo', 'Naati Manae Paya Soup', 'Naati Koli Fry', 'Puliyogare', 'Chicken Donne Biryani', 'Naati Koli Biryani', 'Mutton Donne Biryani', 'Naati Manae Soppina Saru', 'Naati Koli Saru', 'Veg Dosa Combo', 'Veg Chapati Combo', 'Veg Coin Paratha Combo', 'Veg Chitranna Combo', 'Veg Puliyogare Combo', 'Veggie Snack Combo', 'Veg Mudde Combo', 'Veg Mini Meal Combo', 'Veg Nammura Oota Combo', 'Veg Super Combo', 'Non Veg Dosa Combo', 'Non Veg Chapati Combo', 'Non Veg Coin Paratha Combo', 'Non Veg Mudde Oota Combo', 'Non Veg Nammura Oota Combo', 'Chicken Biryani Combo', 'Handi Biryani Combo', 'Chicken

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Starter Combo', 'Nammura Mutton Combo', 'Mutton Starter Combo', 'Mangalore Fish
Fry and Fish Curry Rice Combo', 'Mangalore Fish Curry Rice Mini Combo', 'Naati
Manae Paya Soup', 'Assorted Bajjis', 'Gobi Manchurian', 'Gobi Majestic Fry',
'Gobi 65', 'Naati Style Onion Pakoda', 'Onion Rings', 'Chilli Bajji', 'Potato
Bajji', 'Capsicum Bajji', 'Naati Koli Fry', 'Hasi Menasina Koli Fry', 'Kempu
Menasina Koli Fry', 'Chicken Kebab', 'Chicken Lollipop', 'Boneless Chicken
Sholay Kebab', 'Naati Manae Special Chicken Platter', 'Nammura Mutton Fry',
'Mutton Liver Fry', 'Mutton Boti Fry', 'Mutter Pepper Fry', 'Naati Manae Soppina
Saru', 'Naati Manae Sambar', 'Naati Manae Traditional Rasam', 'Naati Manae Veg
Kurma', 'Chicken Chops Gravy', 'Naati Koli Saru', 'Mutton Chops Gravy',
'Mangalore Fish Curry', 'Chapati', 'Coin Paratha', 'Chitranna', 'Puliyogare',
'Jeera Rice', 'Mosuranna', 'Veg Dum Biryani', 'Egg Biryani', 'Chicken Donne
Biryani', 'Naati Koli Biryani', 'Chicken Hasi Menasina Donne Biryani', 'Kempu
Menasina Donne Biryani', 'Boneless Chicken Handi Biryani', 'Chicken Handi
Biryani', 'Kebab and Biryani', 'Lollipop Biryani', 'Biryani Rice', 'Mutton Donne
Biryani', 'Naati Manae Special Dosa', 'Raagi Ball', 'Karnataka Traditional
Buttermilk']
1
Name: menu_item, Length: 9098, dtype: int64
```

[354]:
```
df1.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 51717 entries, 0 to 51716
Data columns (total 17 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   url                          51717 non-null  object
 1   address                      51717 non-null  object
 2   name                         51717 non-null  object
 3   online_order                 51717 non-null  object
 4   book_table                   51717 non-null  object
 5   rate                         43942 non-null  object
 6   votes                        51717 non-null  int64
 7   phone                        50509 non-null  object
 8   location                     51696 non-null  object
 9   rest_type                    51490 non-null  object
 10  dish_liked                   23639 non-null  object
 11  cuisines                     51672 non-null  object
 12  approx_cost(for two people)  51371 non-null  object
 13  reviews_list                 51717 non-null  object
 14  menu_item                    51717 non-null  object
 15  listed_in(type)              51717 non-null  object
 16  listed_in(city)              51717 non-null  object
dtypes: int64(1), object(16)
memory usage: 6.7+ MB
```

[355]:
```
df1.head()
```

[355]:

	url	address	name	online_order	book_table	rate	votes	phone	location	rest_type	dish_liked	cuisines	approx_cost(for two people)	reviews_list	menu_item
0	https://www.zomato.com/bangalore/jalsa-banasha…	942, 21st Main Road, 2nd Stage, Banashankari, …	Jalsa	Yes	Yes	4.1/5	775	080 42297555\r\n+91 9743772233	Banashankari	Casual Dining	Pasta, Lunch Buffet, Masala Papad, Paneer Laja…	North Indian, Mughlai, Chinese	800	[('Rated 4.0', 'RATED\n A beautiful place to …	[]
1	https://www.zomato.com/bangalore/spice-elephan…	2nd Floor, 80 Feet Road, Near Big Bazaar, 6th …	Spice Elephant	Yes	No	4.1/5	787	080 41714161	Banashankari	Casual Dining	Momos, Lunch Buffet, Chocolate Nirvana, Thai G…	Chinese, North Indian, Thai	800	[('Rated 4.0', 'RATED\n Had been here for din…	[]
2	https://www.zomato.com/SanchurroBangalore?cont…	1112, Next to KIMS Medical College, 17th Cross…	San Churro Cafe	Yes	No	3.8/5	918	+91 9663487993	Banashankari	Cafe, Casual Dining	Churros, Cannelloni, Minestrone Soup, Hot Choc…	Cafe, Mexican, Italian	800		
3	https://www.zomato.com/bangalore/addhuri-udupi…	1st Floor, Annakuteera, 3rd Stage, Banashankar…	Addhuri Udupi Bhojana	No	No	3.7/5	88	+91 9620009302	Banashankari	Quick Bites	Masala Dosa	South Indian, North Indian	300		
4	https://www.zomato.com/bangalore/grand-village…	10, 3rd Floor, Lakshmi Associates, Gandhi Baza…	Grand Village	No	No	3.8/5	166	+91 8026612447\r\n+91 9901210005	Basavanagudi	Casual Dining	Panipuri, Gol Gappe	North Indian, Rajasthani	600		

```
2  [('Rated 3.0', "RATED\n  Ambience is not that …        []
3  [('Rated 4.0', "RATED\n  Great food and proper…        []
4  [('Rated 4.0', 'RATED\n  Very good restaurant …        []

  listed_in(type) listed_in(city)
0          Buffet    Banashankari
1          Buffet    Banashankari
2          Buffet    Banashankari
3          Buffet    Banashankari
4          Buffet    Banashankari
```

[356]:
```
df1['listed_in(type)'].value_counts()
```

[356]:
```
Delivery              25942
Dine-out              17779
Desserts               3593
Cafes                  1723
Drinks & nightlife     1101
Buffet                  882
Pubs and bars           697
Name: listed_in(type), dtype: int64
```

0.0.1 Dropping few columns:

[357]:
```
df1.columns
```

[357]:
```
Index(['url', 'address', 'name', 'online_order', 'book_table', 'rate', 'votes',
       'phone', 'location', 'rest_type', 'dish_liked', 'cuisines',
       'approx_cost(for two people)', 'reviews_list', 'menu_item',
       'listed_in(type)', 'listed_in(city)'],
      dtype='object')
```

[358]:
```
df = df1.
↪drop(['address','phone','dish_liked','cuisines','reviews_list','listed_in(city)','reviews_l
↪axis = 1)
```

[359]:
```
df.head()
```

[359]:
```
                                                 url                   name  \
0  https://www.zomato.com/bangalore/jalsa-banasha…                  Jalsa
1  https://www.zomato.com/bangalore/spice-elephan…         Spice Elephant
2  https://www.zomato.com/SanchurroBangalore?cont…        San Churro Cafe
3  https://www.zomato.com/bangalore/addhuri-udupi…  Addhuri Udupi Bhojana
4  https://www.zomato.com/bangalore/grand-village…          Grand Village

  online_order book_table   rate  votes      location            rest_type  \
0          Yes        Yes  4.1/5    775  Banashankari        Casual Dining
```

```
1          Yes         No  4.1/5    787  Banashankari        Casual Dining
2          Yes         No  3.8/5    918  Banashankari  Cafe, Casual Dining
3           No         No  3.7/5     88  Banashankari          Quick Bites
4           No         No  3.8/5    166  Basavanagudi        Casual Dining

  approx_cost(for two people) listed_in(type)
0                         800          Buffet
1                         800          Buffet
2                         800          Buffet
3                         300          Buffet
4                         600          Buffet
```

[360]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 51717 entries, 0 to 51716
Data columns (total 10 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   url                          51717 non-null  object
 1   name                         51717 non-null  object
 2   online_order                 51717 non-null  object
 3   book_table                   51717 non-null  object
 4   rate                         43942 non-null  object
 5   votes                        51717 non-null  int64
 6   location                     51696 non-null  object
 7   rest_type                    51490 non-null  object
 8   approx_cost(for two people)  51371 non-null  object
 9   listed_in(type)              51717 non-null  object
dtypes: int64(1), object(9)
memory usage: 3.9+ MB
```

- Observations :
- Rating is in string,
- approx cost is also in string

[361]:
```
df['rate'].unique()
```

[361]:
```
array(['4.1/5', '3.8/5', '3.7/5', '3.6/5', '4.6/5', '4.0/5', '4.2/5',
       '3.9/5', '3.1/5', '3.0/5', '3.2/5', '3.3/5', '2.8/5', '4.4/5',
       '4.3/5', 'NEW', '2.9/5', '3.5/5', nan, '2.6/5', '3.8 /5', '3.4/5',
       '4.5/5', '2.5/5', '2.7/5', '4.7/5', '2.4/5', '2.2/5', '2.3/5',
       '3.4 /5', '-', '3.6 /5', '4.8/5', '3.9 /5', '4.2 /5', '4.0 /5',
       '4.1 /5', '3.7 /5', '3.1 /5', '2.9 /5', '3.3 /5', '2.8 /5',
       '3.5 /5', '2.7 /5', '2.5 /5', '3.2 /5', '2.6 /5', '4.5 /5',
       '4.3 /5', '4.4 /5', '4.9/5', '2.1/5', '2.0/5', '1.8/5', '4.6 /5',
       '4.9 /5', '3.0 /5', '4.8 /5', '2.3 /5', '4.7 /5', '2.4 /5',
       '2.1 /5', '2.2 /5', '2.0 /5', '1.8 /5'], dtype=object)
```

[362]:
```
# there is NEW, -  lets take care of this
# Note in pandas if there is missing data it will shows as nan and it will␣
 ↪ignore that from computation, dont show any error. . skip NaNs by default
def cleaning_ratings(rating):
    if(rating == 'NEW' or rating == '-'):
        return np.nan
    else:
        return float(str(rating).split('/')[0])
```

[363]:
```
df['rate']
```

[363]:
```
0         4.1/5
1         4.1/5
2         3.8/5
3         3.7/5
4         3.8/5
          …
51712    3.6 /5
51713       NaN
51714       NaN
51715    4.3 /5
51716    3.4 /5
Name: rate, Length: 51717, dtype: object
```

[364]:
```
df['rate'].apply(cleaning_ratings)
```

[364]:
```
0        4.1
1        4.1
2        3.8
3        3.7
4        3.8
        …
51712    3.6
51713    NaN
51714    NaN
51715    4.3
51716    3.4
Name: rate, Length: 51717, dtype: float64
```

[365]:
```
#Redefine rate
df['rate'] = df['rate'].apply(cleaning_ratings)
df.head()
```

[365]:
```
                                                 url                   name  \
0  https://www.zomato.com/bangalore/jalsa-banasha…                  Jalsa
1  https://www.zomato.com/bangalore/spice-elephan…         Spice Elephant
2  https://www.zomato.com/SanchurroBangalore?cont…        San Churro Cafe
```

```
3  https://www.zomato.com/bangalore/addhuri-udupi…  Addhuri Udupi Bhojana
4  https://www.zomato.com/bangalore/grand-village…          Grand Village

  online_order book_table  rate  votes      location            rest_type  \
0          Yes        Yes   4.1    775  Banashankari        Casual Dining
1          Yes         No   4.1    787  Banashankari        Casual Dining
2          Yes         No   3.8    918  Banashankari  Cafe, Casual Dining
3           No         No   3.7     88  Banashankari          Quick Bites
4           No         No   3.8    166  Basavanagudi        Casual Dining

  approx_cost(for two people) listed_in(type)
0                         800          Buffet
1                         800          Buffet
2                         800          Buffet
3                         300          Buffet
4                         600          Buffet
```

[366]:
```
# Checking how many null values (missing values) are there:
df.isnull().sum()
```

[366]:
```
url                            0
name                           0
online_order                   0
book_table                     0
rate                       10052
votes                          0
location                      21
rest_type                    227
approx_cost(for two people)  346
listed_in(type)                0
dtype: int64
```

[367]:
```
np.round(df['rate'].mean(),1)
```

[367]:
```
3.7
```

0.0.2 Filling the missing data in rating with mean ratings:

[368]:
```
df['rate'].fillna(np.round(df['rate'].mean(),1))
```

[368]:
```
0        4.1
1        4.1
2        3.8
3        3.7
4        3.8
        …
51712    3.6
```

```
51713    3.7
51714    3.7
51715    4.3
51716    3.4
Name: rate, Length: 51717, dtype: float64
```

[369]:
```
df['rate'] = df['rate'].fillna(np.round(df['rate'].mean(),1))
df['rate'].isnull().sum()
```

[369]:
```
0
```

[370]:
```
df.head()
```

[370]:

	url	name	online_order	book_table	rate	votes	location	rest_type	approx_cost(for two people)	listed_in(type)
0	https://www.zomato.com/bangalore/jalsa-banasha…	Jalsa	Yes	Yes	4.1	775	Banashankari	Casual Dining	800	Buffet
1	https://www.zomato.com/bangalore/spice-elephan…	Spice Elephant	Yes	No	4.1	787	Banashankari	Casual Dining	800	Buffet
2	https://www.zomato.com/SanchurroBangalore?cont…	San Churro Cafe	Yes	No	3.8	918	Banashankari	Cafe, Casual Dining	800	Buffet
3	https://www.zomato.com/bangalore/addhuri-udupi…	Addhuri Udupi Bhojana	No	No	3.7	88	Banashankari	Quick Bites	300	Buffet
4	https://www.zomato.com/bangalore/grand-village…	Grand Village	No	No	3.8	166	Basavanagudi	Casual Dining	600	Buffet

[371]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 51717 entries, 0 to 51716
Data columns (total 10 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   url                          51717 non-null  object
 1   name                         51717 non-null  object
 2   online_order                 51717 non-null  object
 3   book_table                   51717 non-null  object
 4   rate                         51717 non-null  float64
 5   votes                        51717 non-null  int64
```

```
 6   location                     51696 non-null  object
 7   rest_type                    51490 non-null  object
 8   approx_cost(for two people)  51371 non-null  object
 9   listed_in(type)              51717 non-null  object
dtypes: float64(1), int64(1), object(8)
memory usage: 3.9+ MB
```

[372]:
```
#  approx_cost(for two people)  51371 non-null  object  : we need to fix this
```

[373]:
```
df['approx_cost(for two people)'].unique()
```

[373]:
```
array(['800', '300', '600', '700', '550', '500', '450', '650', '400',
       '900', '200', '750', '150', '850', '100', '1,200', '350', '250',
       '950', '1,000', '1,500', '1,300', '199', '80', '1,100', '160',
       '1,600', '230', '130', '50', '190', '1,700', nan, '1,400', '180',
       '1,350', '2,200', '2,000', '1,800', '1,900', '330', '2,500',
       '2,100', '3,000', '2,800', '3,400', '40', '1,250', '3,500',
       '4,000', '2,400', '2,600', '120', '1,450', '469', '70', '3,200',
       '60', '560', '240', '360', '6,000', '1,050', '2,300', '4,100',
       '5,000', '3,700', '1,650', '2,700', '4,500', '140'], dtype=object)
```

[374]:
```
# int('1,200'.replace(',','') )This is the logic we do, since we can convert␣
 ↪directly to float/int
```

[375]:
```
#df['approx_cost(for two people)'].isnull().sum()
df.isnull().sum()
```

[375]:
```
url                             0
name                            0
online_order                    0
book_table                      0
rate                            0
votes                           0
location                       21
rest_type                     227
approx_cost(for two people)   346
listed_in(type)                 0
dtype: int64
```

- 346 null values :
- And in total approx ~600 rown at max in total, If i drop all those rows it wont affect, because my data set length is alose to 50k.

[376]:
```
# Hence we can remove thows rows:
df.dropna()
```

[376]:

	url	\
0	https://www.zomato.com/bangalore/jalsa-banasha…	
1	https://www.zomato.com/bangalore/spice-elephan…	
2	https://www.zomato.com/SanchurroBangalore?cont…	
3	https://www.zomato.com/bangalore/addhuri-udupi…	
4	https://www.zomato.com/bangalore/grand-village…	
…	…	
51712	https://www.zomato.com/bangalore/best-brews-fo…	
51713	https://www.zomato.com/bangalore/vinod-bar-and…	
51714	https://www.zomato.com/bangalore/plunge-sherat…	
51715	https://www.zomato.com/bangalore/chime-sherato…	
51716	https://www.zomato.com/bangalore/the-nest-the-…	

	name	online_order	\
0	Jalsa	Yes	
1	Spice Elephant	Yes	
2	San Churro Cafe	Yes	
3	Addhuri Udupi Bhojana	No	
4	Grand Village	No	
…	…	…	
51712	Best Brews - Four Points by Sheraton Bengaluru…	No	
51713	Vinod Bar And Restaurant	No	
51714	Plunge - Sheraton Grand Bengaluru Whitefield H…	No	
51715	Chime - Sheraton Grand Bengaluru Whitefield Ho…	No	
51716	The Nest - The Den Bengaluru	No	

	book_table	rate	votes	location	\
0	Yes	4.1	775	Banashankari	
1	No	4.1	787	Banashankari	
2	No	3.8	918	Banashankari	
3	No	3.7	88	Banashankari	
4	No	3.8	166	Basavanagudi	
…	…	…	…	…	
51712	No	3.6	27	Whitefield	
51713	No	3.7	0	Whitefield	
51714	No	3.7	0	Whitefield	
51715	Yes	4.3	236	ITPL Main Road, Whitefield	
51716	No	3.4	13	ITPL Main Road, Whitefield	

	rest_type	approx_cost(for two people)	listed_in(type)
0	Casual Dining	800	Buffet
1	Casual Dining	800	Buffet
2	Cafe, Casual Dining	800	Buffet
3	Quick Bites	300	Buffet
4	Casual Dining	600	Buffet
…	…	…	…
51712	Bar	1,500	Pubs and bars

```
51713                  Bar                          600   Pubs and bars
51714                  Bar                        2,000   Pubs and bars
51715                  Bar                        2,500   Pubs and bars
51716   Bar, Casual Dining                        1,500   Pubs and bars

[51167 rows x 10 columns]
```

[377]:
```
# Assigning it to df
df = df.dropna()
```

- We are losing 1% of data

[378]:
```
df.isnull().sum()
```

[378]:
```
url                            0
name                           0
online_order                   0
book_table                     0
rate                           0
votes                          0
location                       0
rest_type                      0
approx_cost(for two people)    0
listed_in(type)                0
dtype: int64
```

[379]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 51167 entries, 0 to 51716
Data columns (total 10 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   url                          51167 non-null  object
 1   name                         51167 non-null  object
 2   online_order                 51167 non-null  object
 3   book_table                   51167 non-null  object
 4   rate                         51167 non-null  float64
 5   votes                        51167 non-null  int64
 6   location                     51167 non-null  object
 7   rest_type                    51167 non-null  object
 8   approx_cost(for two people)  51167 non-null  object
 9   listed_in(type)              51167 non-null  object
dtypes: float64(1), int64(1), object(8)
memory usage: 4.3+ MB
```

- Now there is no null valuesin df

[380]:
```
def cleaned_approx_cost(value):
    value = str(value)
    if ',' in value:
        return float(value.replace(',',''))
    else:
        return float(value)
```

[381]:
```
df['approx_cost(for two people)'].apply(cleaned_approx_cost)
```

[381]:
```
0         800.0
1         800.0
2         800.0
3         300.0
4         600.0
          ...
51712    1500.0
51713     600.0
51714    2000.0
51715    2500.0
51716    1500.0
Name: approx_cost(for two people), Length: 51167, dtype: float64
```

[382]:
```
# reassign it to df
df['approx_cost(for two people)']  = df['approx_cost(for two people)'].
 ↪apply(cleaned_approx_cost)
```

[383]:
```
# Checking for duplicated rows
df.duplicated().sum()
```

[383]:
```
0
```

[384]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 51167 entries, 0 to 51716
Data columns (total 10 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   url                          51167 non-null  object
 1   name                         51167 non-null  object
 2   online_order                 51167 non-null  object
 3   book_table                   51167 non-null  object
 4   rate                         51167 non-null  float64
 5   votes                        51167 non-null  int64
 6   location                     51167 non-null  object
 7   rest_type                    51167 non-null  object
```

```
 8   approx_cost(for two people)  51167 non-null  float64
 9   listed_in(type)              51167 non-null  object
dtypes: float64(2), int64(1), object(7)
memory usage: 4.3+ MB
```

0.0.3 Understand this rest_type column

[385]:
```
df['rest_type'].value_counts()
```

[385]:
```
Quick Bites                   19048
Casual Dining                 10275
Cafe                           3687
Delivery                       2587
Dessert Parlor                 2245
                              ...
Dessert Parlor, Kiosk             2
Dessert Parlor, Food Court        2
Pop Up                            2
Sweet Shop, Dessert Parlor        1
Quick Bites, Kiosk                1
Name: rest_type, Length: 93, dtype: int64
```

[386]:
```
rest_type = df['rest_type'].value_counts()
rest_type
#rest_type[rest_type < 450]
```

[386]:
```
Quick Bites                   19048
Casual Dining                 10275
Cafe                           3687
Delivery                       2587
Dessert Parlor                 2245
                              ...
Dessert Parlor, Kiosk             2
Dessert Parlor, Food Court        2
Pop Up                            2
Sweet Shop, Dessert Parlor        1
Quick Bites, Kiosk                1
Name: rest_type, Length: 93, dtype: int64
```

[387]:
```
rest_type[rest_type < 450]
```

[387]:
```
Bar, Casual Dining            415
Lounge                        396
Pub                           357
Fine Dining                   346
Casual Dining, Cafe           311
                             ...
```

```
Dessert Parlor, Kiosk           2
Dessert Parlor, Food Court      2
Pop Up                          2
Sweet Shop, Dessert Parlor      1
Quick Bites, Kiosk              1
Name: rest_type, Length: 81, dtype: int64
```

[388]:
```
type_below_500 = rest_type[rest_type < 500]
```

[389]:
```
def clean_rest_type(value):
    if (value) in type_below_500:
        return 'Others'
    else:
        return value
```

[390]:
```
df['rest_type'].value_counts()
```

[390]:
```
Quick Bites                   19048
Casual Dining                 10275
Cafe                           3687
Delivery                       2587
Dessert Parlor                 2245
                              ...
Dessert Parlor, Kiosk             2
Dessert Parlor, Food Court        2
Pop Up                            2
Sweet Shop, Dessert Parlor        1
Quick Bites, Kiosk                1
Name: rest_type, Length: 93, dtype: int64
```

[391]:
```
clean_rest_type('Quick Bites')
```

[391]:
```
'Quick Bites'
```

[392]:
```
clean_rest_type('Quick Bites, Kiosk')
```

[392]:
```
'Others'
```

[393]:
```
df['rest_type'].apply(clean_rest_type)
```

[393]:
```
0        Casual Dining
1        Casual Dining
2               Others
3          Quick Bites
4        Casual Dining
             ...
51712              Bar
```

```
51713              Bar
51714              Bar
51715              Bar
51716           Others
Name: rest_type, Length: 51167, dtype: object
```

[394]:
```
# Reassign
df['rest_type'] = df['rest_type'].apply(clean_rest_type)
```

[395]:
```
df['rest_type'].value_counts()
```

[395]:
```
Quick Bites           19048
Casual Dining         10275
Others                 6860
Cafe                   3687
Delivery               2587
Dessert Parlor         2245
Takeaway, Delivery     2016
Bakery                 1141
Casual Dining, Bar     1136
Beverage Shop           867
Bar                     686
Food Court              619
Name: rest_type, dtype: int64
```

[396]:
```
df['location'].value_counts()
#len(df['location'].value_counts())
```

[396]:
```
BTM                    5071
HSR                    2496
Koramangala 5th Block  2481
JP Nagar               2219
Whitefield             2117
                       …
West Bangalore            6
Yelahanka                 5
Jakkur                    3
Rajarajeshwari Nagar      2
Peenya                    1
Name: location, Length: 93, dtype: int64
```

[397]:
```
rest_loc = df['location'].value_counts()
rest_loc[rest_loc < 500]
```

[397]:
```
Shivajinagar                  499
Cunningham Road               491
Domlur                        482
```

Old Airport Road	437
Ejipura	434
Commercial Street	370
St. Marks Road	343
Koramangala 8th Block	294
Vasanth Nagar	293
Jeevan Bhima Nagar	268
Wilson Garden	246
Bommanahalli	236
Koramangala 3rd Block	216
Kumaraswamy Layout	194
Thippasandra	191
Basaveshwara Nagar	187
Nagawara	187
Seshadripuram	165
Hennur	159
Majestic	155
HBR Layout	153
Infantry Road	151
Race Course Road	139
City Market	122
Yeshwantpur	119
ITPL Main Road, Whitefield	113
Varthur Main Road, Whitefield	109
South Bangalore	107
Koramangala 2nd Block	102
Hosur Road	102
Kaggadasapura	101
CV Raman Nagar	90
RT Nagar	78
Vijay Nagar	78
Sanjay Nagar	76
Sadashiv Nagar	63
Sahakara Nagar	53
Koramangala	48
East Bangalore	43
Jalahalli	38
Magadi Road	34
Rammurthy Nagar	32
Sankey Road	27
Langford Town	27
Mysore Road	22
Old Madras Road	22
Kanakapura Road	19
KR Puram	18
Uttarahalli	17
Hebbal	14

```
North Bangalore                  14
Nagarbhavi                        9
Kengeri                           9
Central Bangalore                 8
West Bangalore                    6
Yelahanka                         5
Jakkur                            3
Rajarajeshwari Nagar              2
Peenya                            1
Name: location, dtype: int64
```

[398]:
```
loc_below_500 = rest_loc[rest_loc < 500]
```

[399]:
```
def clean_rest_loc(loc):
    if loc in loc_below_500:
        return 'Others'
    else:
        return loc
```

[400]:
```
df['location'].apply(clean_rest_loc)
```

[400]:
```
0        Banashankari
1        Banashankari
2        Banashankari
3        Banashankari
4        Basavanagudi
             ...
51712      Whitefield
51713      Whitefield
51714      Whitefield
51715          Others
51716          Others
Name: location, Length: 51167, dtype: object
```

[401]:
```
df['location'] = df['location'].apply(clean_rest_loc)
df['location'].value_counts()
```

[401]:
```
Others                   8021
BTM                      5071
HSR                      2496
Koramangala 5th Block    2481
JP Nagar                 2219
Whitefield               2117
Indiranagar              2033
Jayanagar                1916
Marathahalli             1811
Bannerghatta Road        1611
```

```
Bellandur                 1271
Electronic City           1249
Koramangala 1st Block     1237
Brigade Road              1218
Koramangala 7th Block     1176
Koramangala 6th Block     1129
Sarjapur Road             1049
Koramangala 4th Block     1017
Ulsoor                    1017
Banashankari               906
MG Road                    894
Kalyan Nagar               843
Richmond Road              804
Malleshwaram               724
Frazer Town                720
Basavanagudi               684
Residency Road             674
Brookefield                656
Banaswadi                  645
New BEL Road               644
Kammanahalli               640
Rajajinagar                591
Church Street              569
Lavelle Road               523
Shanti Nagar               511
Name: location, dtype: int64
```

[402]:
```
df.head()
```

[402]:

	url	name	online_order	book_table	rate	votes	location	rest_type	approx_cost(for two people)	listed_in(type)
0	https://www.zomato.com/bangalore/jalsa-banasha…	Jalsa	Yes	Yes	4.1	775	Banashankari	Casual Dining	800.0	Buffet
1	https://www.zomato.com/bangalore/spice-elephan…	Spice Elephant	Yes	No	4.1	787	Banashankari	Casual Dining	800.0	Buffet
2	https://www.zomato.com/SanchurroBangalore?cont…	San Churro Cafe	Yes	No	3.8	918	Banashankari	Others	800.0	Buffet
3	https://www.zomato.com/bangalore/addhuri-udupi…	Addhuri Udupi Bhojana	No	No	3.7	88	Banashankari	Quick Bites		
4	https://www.zomato.com/bangalore/grand-village…	Grand Village	No	No	3.8	166	Basavanagudi	Casual Dining		

```
3                       300.0          Buffet
4                       600.0          Buffet
```

[403]:
```
df['listed_in(type)'].value_counts()
```

[403]:
```
Delivery              25669
Dine-out              17586
Desserts               3559
Cafes                  1703
Drinks & nightlife     1091
Buffet                  871
Pubs and bars           688
Name: listed_in(type), dtype: int64
```

0.0.4 Number of Resturants delivers online and offline order as per Zomato dataset:

[404]:
```
df['online_order']
```

[404]:
```
0        Yes
1        Yes
2        Yes
3         No
4         No
        ...
51712     No
51713     No
51714     No
51715     No
51716     No
Name: online_order, Length: 51167, dtype: object
```

[405]:
```
df['online_order'].value_counts()
```

[405]:
```
Yes    30327
No     20840
Name: online_order, dtype: int64
```

[406]:
```
print(f"There are  {df['online_order'].value_counts().iloc[0]} number of␣
 ↪resturants delivers order online in bangalore as per this Zomato data set.")
```

```
There are  30327 number of resturants delivers order online in bangalore as per
this Zomato data set.
```

[407]:
```
print(f"There are  {df['online_order'].value_counts().iloc[1]} number of␣
 ↪resturants delivers order offline in bangalore as per this Zomato data set.")
```

There are 20840 number of resturants delivers order offline in bangalore as per this Zomato data set.

[ ]:

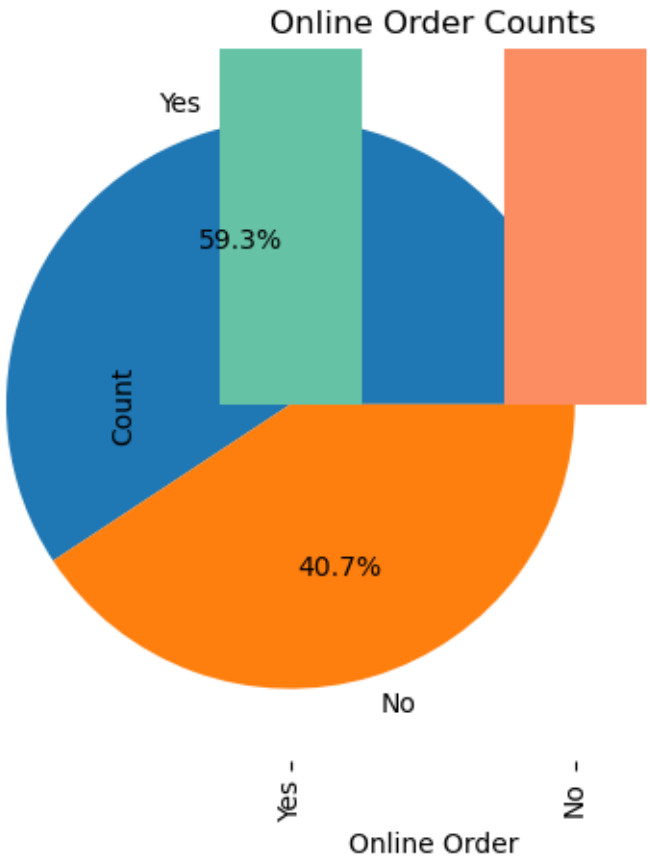
[408]:
```
# Count the values
order_counts = df['online_order'].value_counts()
# plot usimg pandas library
# Plot with labels
order_counts.plot(
    kind='pie',
    labels=order_counts.index,    # These are your labels (e.g., 'Yes', 'No')
    autopct='%1.1f%%',            # Show percentage with 1 decimal
    ylabel='',                    # Remove y-label
    title='Online Order Distribution'
)
```

[408]:
```
<Axes: title={'center': 'Online Order Distribution'}>
```

[409]:
```
df['online_order'].value_counts().plot(kind = 'bar', color = sns.
 ↪color_palette())
```

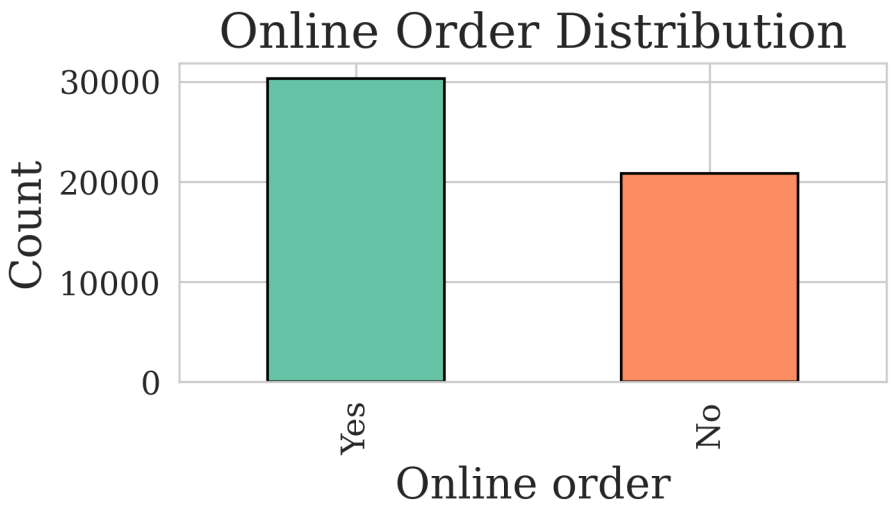
[409]:
```
<Axes: title={'center': 'Online Order Distribution'}>
```

[410]:
```
# Get a palette of colors
colors = sns.color_palette("Set2")  # or "pastel", "muted", etc.

# Bar plot with Seaborn colors
df['online_order'].value_counts().plot(kind='bar', color=colors)
plt.title('Online Order Counts')
plt.ylabel('Count')
plt.xlabel('Online Order')
plt.show()
```



[411]:
```
import matplotlib.pyplot as plt
import seaborn as sns

# Set journal-style aesthetics
sns.set_style("whitegrid")
plt.rcParams.update({
    'font.family': 'serif',
    'font.serif': ['DejaVu Serif'],
    'font.size': 14,
    'axes.labelsize': 16,
    'axes.titlesize': 18,
    'xtick.labelsize': 12,
    'ytick.labelsize': 12,
    'legend.fontsize': 12,
    'figure.dpi': 300
})

# Set figure size here
plt.figure(figsize=(5, 3))  # width=5 inches, height=3 inches
```

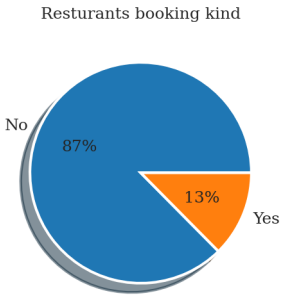
```
# Plot
df['online_order'].value_counts().plot(
    kind='bar',
    color=sns.color_palette('Set2'),
    edgecolor='black'
)

# Labels and title
plt.title('Online Order Distribution', fontsize=18)
plt.xlabel('Online order', fontsize=16)
plt.ylabel('Count', fontsize=16)
plt.tight_layout()

# Show
plt.show()
```



[412]:
```
table_counts = df['book_table'].value_counts()
```

[413]:
```
plt.figure(figsize=(2, 2))  # Very small figure

plt.pie(
    table_counts.values,
    labels=table_counts.index,
    textprops={'fontsize': 6},       #  Comma added here
    autopct='%1.0f%%',
    shadow=True
)
```

```
plt.title('Resturants booking kind', fontsize=6)  # Small title to fit tiny␣
↪figure
plt.tight_layout()
plt.show()
```



0.0.5 Table booking vs Rate

[414]:
```
df['book_table'] == 'Yes'
```

[414]:
```
0         True
1        False
2        False
3        False
4        False
         …
51712    False
51713    False
51714    False
51715     True
51716    False
Name: book_table, Length: 51167, dtype: bool
```

[415]:
```
df[df['book_table'] == 'Yes'] # Shows the data frame
```

[415]:
```
                                                    url  \
0      https://www.zomato.com/bangalore/jalsa-banasha…
7      https://www.zomato.com/bangalore/onesta-banash…
11     https://www.zomato.com/bangalore/cafe-shuffle-…
12     https://www.zomato.com/bangalore/the-coffee-sh…
44     https://www.zomato.com/bangalore/onesta-banash…
…                                                    …
51701  https://www.zomato.com/bangalore/the-beer-cafe…
51703  https://www.zomato.com/bangalore/olivers-pub-d…
51704  https://www.zomato.com/bangalore/smaaash-white…
```

```
51705  https://www.zomato.com/bangalore/izakaya-gastr…
51715  https://www.zomato.com/bangalore/chime-sherato…

                                                    name online_order  \
0                                                  Jalsa          Yes
7                                                 Onesta          Yes
11                                          Cafe Shuffle          Yes
12                                        The Coffee Shack          Yes
44                                                Onesta          Yes
…                                                    …          …
51701                                     The Beer Cafe          Yes
51703                                 Oliver's Pub & Diner          Yes
51704                                             Smaaash           No
51705                                   Izakaya Gastro Pub          Yes
51715  Chime - Sheraton Grand Bengaluru Whitefield Ho…           No

      book_table  rate  votes      location      rest_type  \
0            Yes   4.1    775  Banashankari  Casual Dining
7            Yes   4.6   2556  Banashankari         Others
11           Yes   4.2    150  Banashankari           Cafe
12           Yes   4.2    164  Banashankari           Cafe
44           Yes   4.6   2556  Banashankari         Others
…          …   …    …           …             …
51701        Yes   4.1    673    Whitefield         Others
51703        Yes   3.9    548    Whitefield         Others
51704        Yes   4.0    189    Whitefield         Others
51705        Yes   3.8    128    Whitefield         Others
51715        Yes   4.3    236        Others            Bar

       approx_cost(for two people) listed_in(type)
0                            800.0          Buffet
7                            600.0           Cafes
11                           600.0           Cafes
12                           500.0           Cafes
44                           600.0        Delivery
…                            …              …
51701                       1400.0   Pubs and bars
51703                       1500.0   Pubs and bars
51704                       1500.0   Pubs and bars
51705                       1200.0   Pubs and bars
51715                       2500.0   Pubs and bars

[6449 rows x 10 columns]
```

[416]:
```
# Grab the ratings of the resturant which accepts table booking
np.round(df[df['book_table'] == 'Yes']['rate'].mean(),2)
```

[416]: 4.13

[417]:
```
avg_rate_with_booking = np.round(df[df['book_table'] == 'Yes']['rate'].mean(),2)
avg_rate_without_booking = np.round(df[df['book_table'] == 'No']['rate'].
 ↪mean(),2)
```

[418]:
```
avg_rate_with_booking
```

[418]: 4.13

[419]:
```
avg_rate_without_booking
```

[419]: 3.64

[420]:
```
#You can reset matplotlib settings to default with:
import matplotlib as mpl
mpl.rcParams.update(mpl.rcParamsDefault)
```

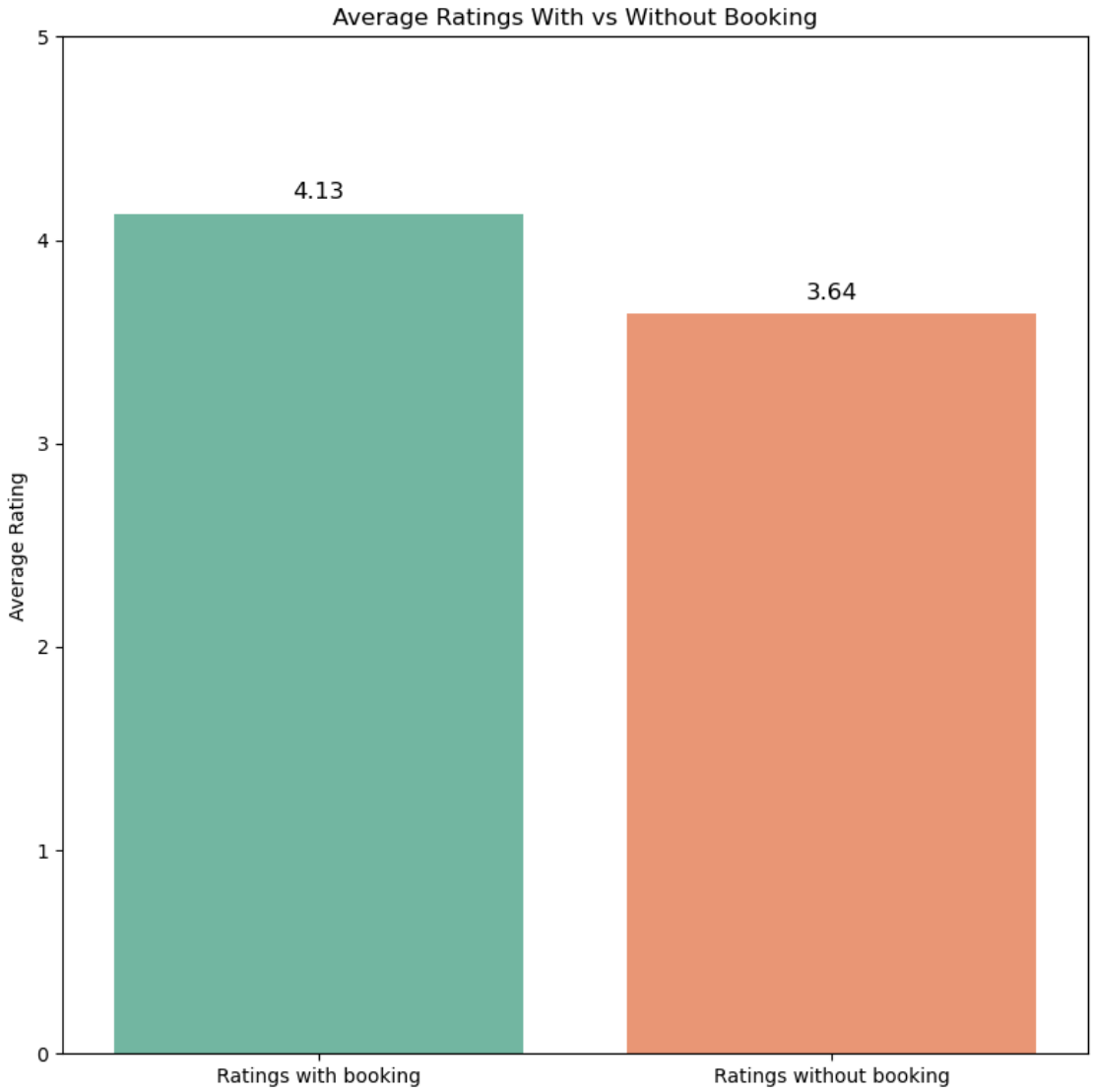
[421]:
```
df2 = pd.DataFrame.from_dict({
    'Ratings with booking' : [avg_rate_with_booking],
    'Ratings without booking' : [avg_rate_without_booking]

})
df2
```

[421]:

	Ratings with booking	Ratings without booking
0	4.13	3.64

[422]:
```
# Reshape your data to long format
df2_melted = df2.melt(var_name='Booking Type', value_name='Average Rating')

# Set figure size
plt.figure(figsize=(8, 8))

# Plot
ax = sns.barplot(
    data=df2_melted,
    x='Booking Type',
    y='Average Rating',
    hue='Booking Type',
    palette='Set2',
    legend=False
)

# Add value labels (containers)
for bar in ax.patches:
    height = bar.get_height()
```

```
    ax.text(
        bar.get_x() + bar.get_width() / 2,
        height + 0.05,  # Slightly above the bar
        f'{height:.2f}',
        ha='center',
        va='bottom',
        fontsize=12
    )

# Titles and labels
plt.title('Average Ratings With vs Without Booking')
plt.ylabel('Average Rating')
plt.xlabel('')
plt.ylim(0, 5)
plt.tight_layout()
plt.show()
```



[423]:
```
df2_melted
```

[423]:
```
              Booking Type  Average Rating
0     Ratings with booking           4.13
1  Ratings without booking           3.64
```

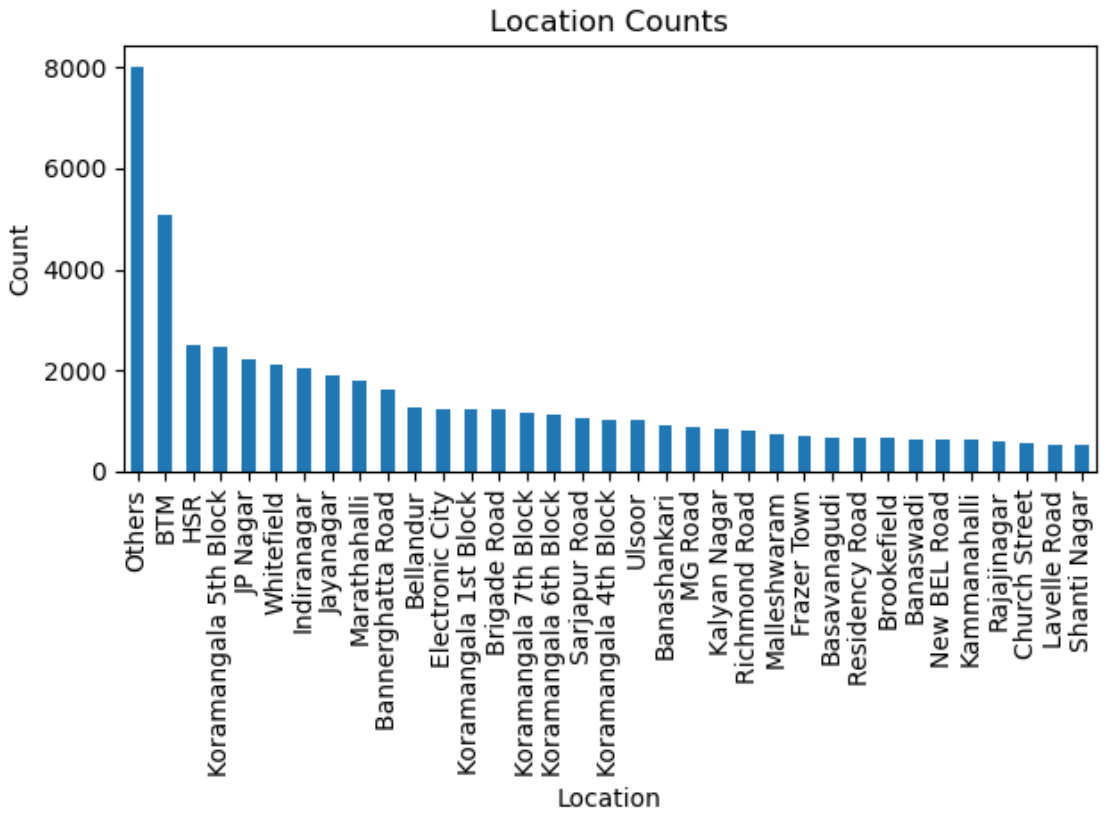
[424]:
```
df.head()
```

[424]:
```
                                                 url                  name  \
0  https://www.zomato.com/bangalore/jalsa-banasha…                 Jalsa
1  https://www.zomato.com/bangalore/spice-elephan…         Spice Elephant
2  https://www.zomato.com/SanchurroBangalore?cont…        San Churro Cafe
3  https://www.zomato.com/bangalore/addhuri-udupi…  Addhuri Udupi Bhojana
```

```
4  https://www.zomato.com/bangalore/grand-village…          Grand Village

  online_order book_table  rate  votes     location      rest_type  \
0          Yes        Yes   4.1    775  Banashankari  Casual Dining
1          Yes         No   4.1    787  Banashankari  Casual Dining
2          Yes         No   3.8    918  Banashankari         Others
3           No         No   3.7     88  Banashankari    Quick Bites
4           No         No   3.8    166  Basavanagudi  Casual Dining

   approx_cost(for two people) listed_in(type)
0                        800.0          Buffet
1                        800.0          Buffet
2                        800.0          Buffet
3                        300.0          Buffet
4                        600.0          Buffet
```

0.0.6 Location with more number of resturants

[425]:
```
import matplotlib.pyplot as plt

df['location'].value_counts().plot(kind='bar')
plt.title('Location Counts')
plt.ylabel('Count')
plt.xlabel('Location')
plt.tight_layout()
plt.show()  # This is what triggers the plot window!
```



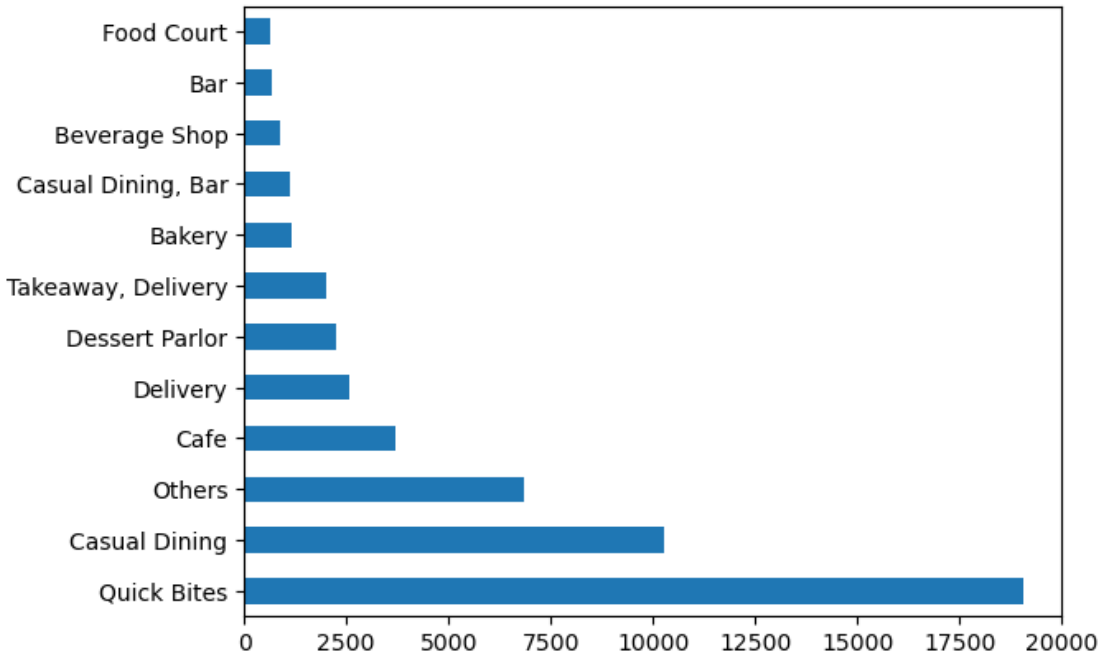
[426]:
```
df.head()
```

[426]:

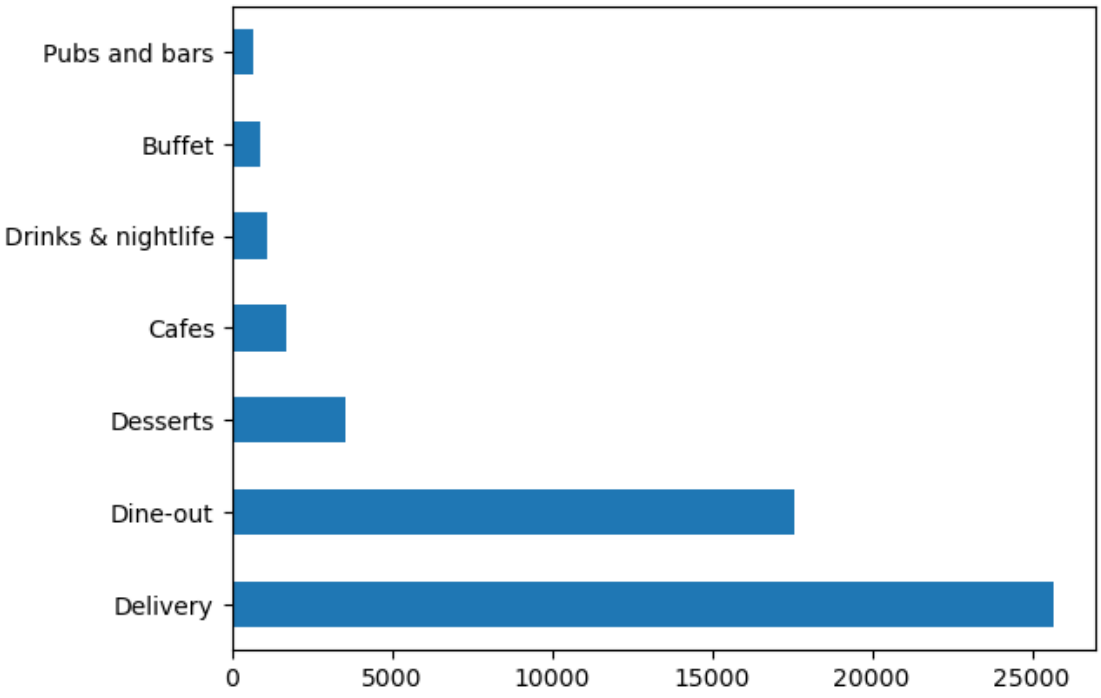
	url	name	online_order	book_table	rate	votes	location	rest_type	approx_cost(for two people)	listed_in(type)
0	https://www.zomato.com/bangalore/jalsa-banasha…	Jalsa	Yes	Yes	4.1	775	Banashankari	Casual Dining	800.0	Buffet
1	https://www.zomato.com/bangalore/spice-elephan…	Spice Elephant	Yes	No	4.1	787	Banashankari	Casual Dining	800.0	Buffet
2	https://www.zomato.com/SanchurroBangalore?cont…	San Churro Cafe	Yes	No	3.8	918	Banashankari	Others	800.0	Buffet
3	https://www.zomato.com/bangalore/addhuri-udupi…	Addhuri Udupi Bhojana	No	No	3.7	88	Banashankari	Quick Bites	300.0	Buffet
4	https://www.zomato.com/bangalore/grand-village…	Grand Village	No	No	3.8	166	Basavanagudi	Casual Dining		

```
4                          600.0           Buffet
```

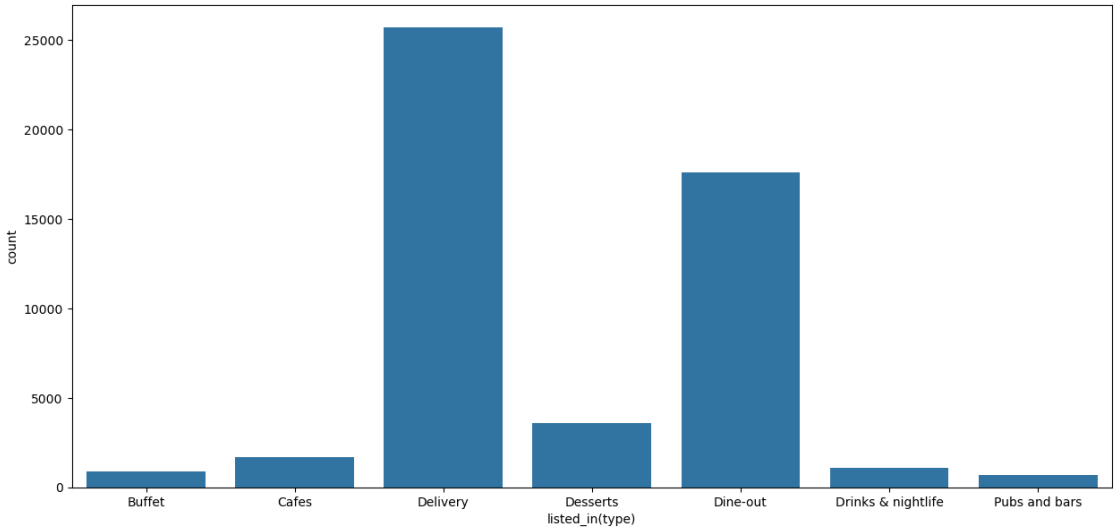
[427]:
```
df['rest_type'].value_counts().plot(kind = 'barh')
plt.show()
```



[428]:
```
df['listed_in(type)'].value_counts().plot(kind = 'barh')
plt.show()
```



[429]:
```
plt.figure(figsize = (15,7))
sns.countplot(x = df['listed_in(type)'])
plt.show()
```



0.1 Cost of Resturants:

[430]:
```
df['approx_cost(for two people)'].value_counts()
```

[430]:
```
300.0     7549
400.0     6525
500.0     4940
200.0     4846
600.0     3693
          …
560.0        1
160.0        1
469.0        1
3700.0       1
5000.0       1
Name: approx_cost(for two people), Length: 70, dtype: int64
```

0.1.1 What is the avg cost of top 10 resturants based on number of chains in bangalore

[451]:
```
# Step 1: Clean cost column (already done)
#df['approx_cost(for two people)'] = (
#    df['approx_cost(for two people)']
 #   .astype(str)
  #  .str.replace(',', '')        # Remove comma
   # .replace('nan', np.nan)
    #.astype(float)
#)

# Step 2: Group cost by restaurant name
res_cost = df['approx_cost(for two people)'].groupby(df['name'])

# Step 3: Get top 10 most frequent restaurant names (chsen based on freq of␣
 ↪appearence)
# Other ways based on ratings, revoew counts etc.
top_names = df['name'].value_counts().head(10).index

# Step 4: Build dictionary with average costs
d1 = {}
for name in top_names:
    d1[name] = np.round(res_cost.get_group(name).mean(), 2)

# Optional: print
print(d1)
```

```
{'Cafe Coffee Day': 844.79, 'Onesta': 600.0, 'Just Bake': 400.0, 'Empire
Restaurant': 685.21, 'Five Star Chicken': 257.86, 'Kanti Sweets': 400.0,
'Petoo': 659.85, 'Polar Bear': 361.54, 'Baskin Robbins': 251.56, 'Pizza Hut':
736.29}
```

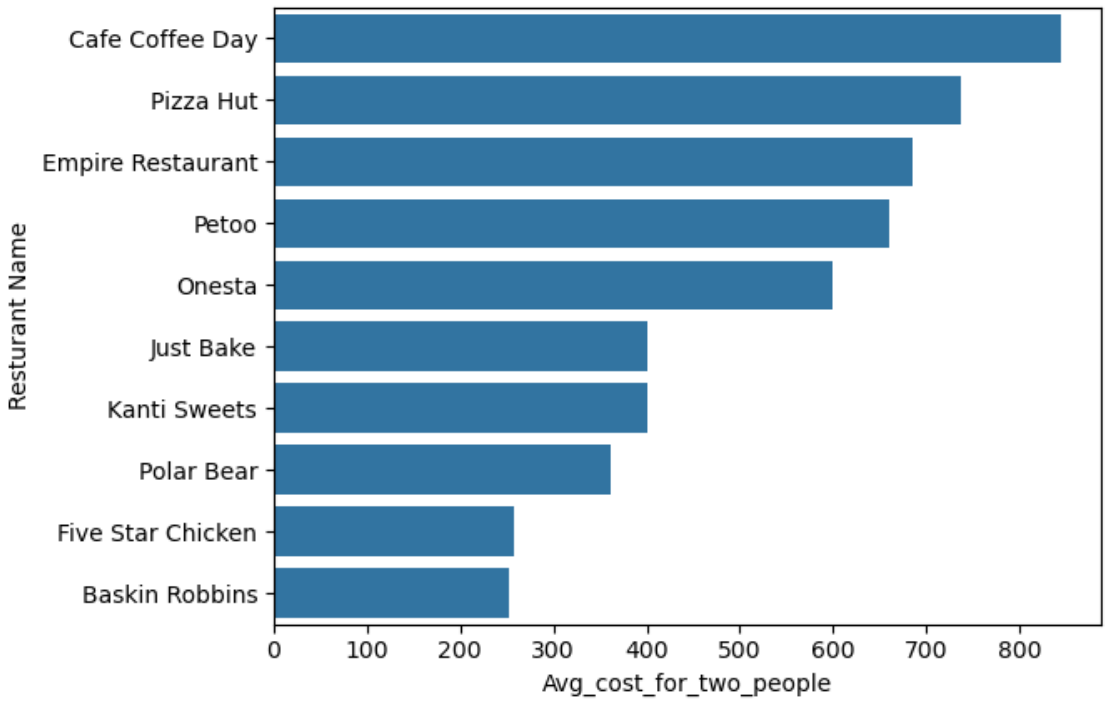
[452]:
```
cost_df = pd.DataFrame(list(d1.items()), columns = ['Resturant Name',␣
↪'Avg_cost_for_two_people'] ).sort_values(by = ['Avg_cost_for_two_people'],␣
↪ascending = False)
```

[453]:
```
cost_df
```

[453]:

	Resturant Name	Avg_cost_for_two_people
0	Cafe Coffee Day	844.79
9	Pizza Hut	736.29
3	Empire Restaurant	685.21
6	Petoo	659.85
1	Onesta	600.00
2	Just Bake	400.00
5	Kanti Sweets	400.00
7	Polar Bear	361.54
4	Five Star Chicken	257.86
8	Baskin Robbins	251.56

[454]:
```
sns.barplot(data = cost_df, x = 'Avg_cost_for_two_people', y = 'Resturant Name')
plt.show()
plt.title('This is avg cost for two people in top 10 resturants based on number␣
↪of branches')
```



[454]:
```
Text(0.5, 1.0, 'This is avg cost for two people in top 10 resturants based on
number of branches')
```

0.1.2 Avg cost based on ratings

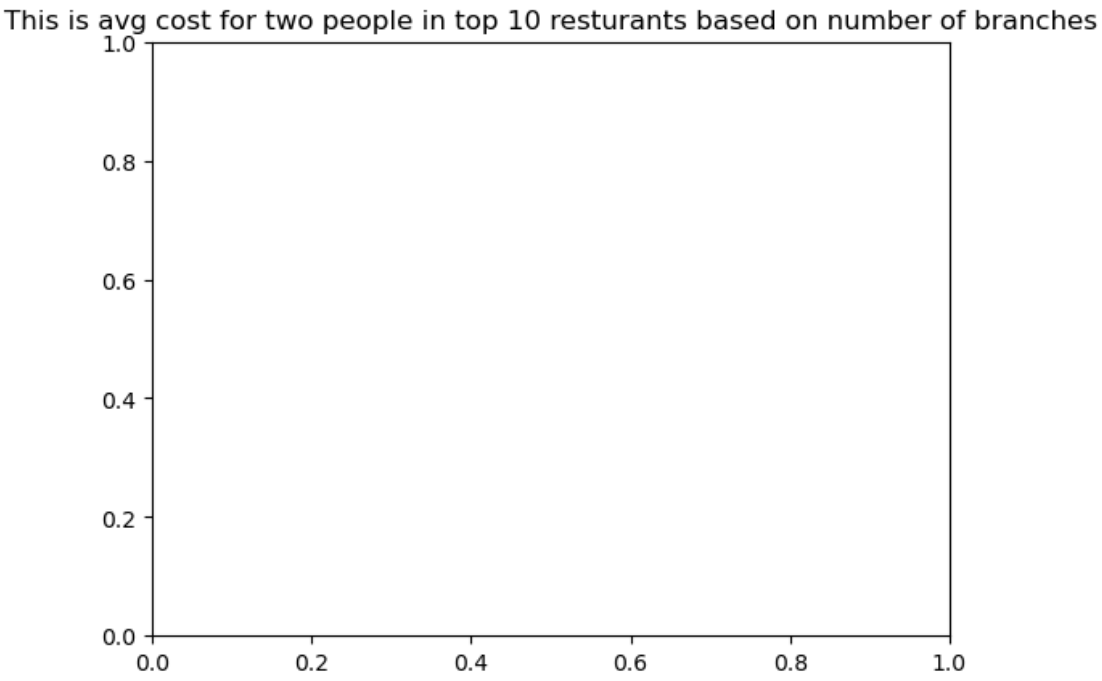
[455]:
```
import pandas as pd
import numpy as np

'''
# Step 1: Clean the 'approx_cost(for two people)' column
df['approx_cost(for two people)'] = (
    df['approx_cost(for two people)']
    .astype(str)
    .str.replace(',', '')
    .replace('nan', np.nan)
    .astype(float)
) '''

# Step 2: Group by restaurant name and compute mean rating and cost
grouped = df.groupby('name').agg({
    'rate': 'mean',
    'approx_cost(for two people)': 'mean'
}).dropna()

# Step 3: Sort by average rating
top10 = grouped.sort_values(by='rate', ascending=False).head(10)

# Step 4: Round the values for clarity
top10 = top10.round(2)

# Step 5: Rename columns (optional)
top10 = top10.rename(columns={
    'rate': 'Average Rating',
    'approx_cost(for two people)': 'Average Cost for Two'
})

print(top10)
```

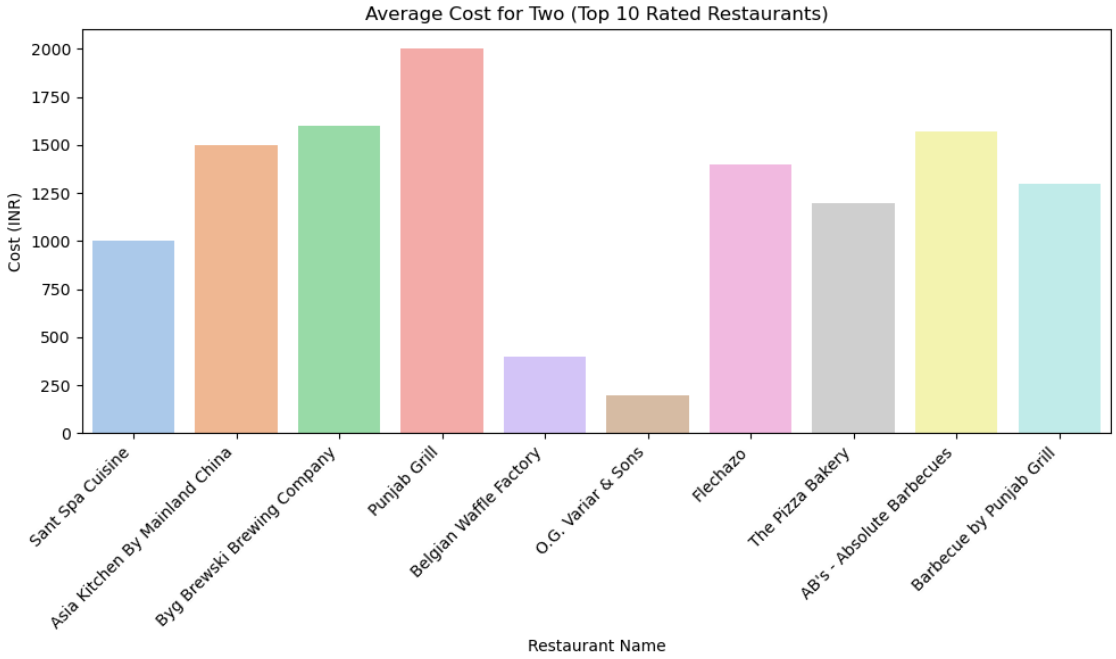
```
                                                  Average Rating  \
name
SantÃ Ã Ã Ã Ã Ã Ã Ã Ã Ã Ã Ã Ã Ã Ã Ã Ã Ã Ã Ã Ã…            4.90
Asia Kitchen By Mainland China                                4.90
Byg Brewski Brewing Company                                   4.90
Punjab Grill                                                  4.87
Belgian Waffle Factory                                        4.84
O.G. Variar & Sons                                            4.80
Flechazo                                                      4.80
```

```
The Pizza Bakery                                          4.80
AB's - Absolute Barbecues                                 4.79
Barbecue by Punjab Grill                                  4.75
```

```
                                                  Average Cost for Two
name
SantÃ Ã Ã Ã Ã Ã Ã Ã Ã Ã Ã Ã Ã Ã Ã Ã Ã Ã Ã Ã Ã…          1000.00
Asia Kitchen By Mainland China                              1500.00
Byg Brewski Brewing Company                                 1600.00
Punjab Grill                                                2000.00
Belgian Waffle Factory                                       400.00
O.G. Variar & Sons                                           200.00
Flechazo                                                    1400.00
The Pizza Bakery                                            1200.00
AB's - Absolute Barbecues                                   1568.42
Barbecue by Punjab Grill                                    1300.00
```

[456]:
```
import matplotlib.pyplot as plt
import seaborn as sns

# Optional: Clean non-ASCII characters from restaurant names
top10 = top10.reset_index()
top10['name'] = top10['name'].str.encode('ascii', errors='ignore').str.
 ↪decode('ascii')

# Plot
plt.figure(figsize=(10, 6))
sns.barplot(
    data=top10,
    x='name',
    y='Average Cost for Two',
    hue='name',               # Avoid FutureWarning
    palette='pastel',
    legend=False              # Prevent duplicated legend
)

plt.title('Average Cost for Two (Top 10 Rated Restaurants)')
plt.xticks(rotation=45, ha='right')
plt.xlabel('Restaurant Name')
plt.ylabel('Cost (INR)')
plt.tight_layout()
plt.show()
```


This is avg cost for two people in top 10 resturants based on number of branches
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0


Average Cost for Two (Top 10 Rated Restaurants)
Cost (INR)
2000
1750
1500
1250
1000
750
500
250
0
Sant Spa Cuisine
Asia Kitchen By Mainland China
Byg Brewski Brewing Company
Punjab Grill
Belgian Waffle Factory
O.G. Variar & Sons
Flechazo
The Pizza Bakery
AB's - Absolute Barbecues
Barbecue by Punjab Grill
Restaurant Name

[457]:
```
df.head()
```

[457]:
```
                                                 url                  name  \
0  https://www.zomato.com/bangalore/jalsa-banasha…                 Jalsa
1  https://www.zomato.com/bangalore/spice-elephan…          Spice Elephant
2  https://www.zomato.com/SanchurroBangalore?cont…         San Churro Cafe
3  https://www.zomato.com/bangalore/addhuri-udupi…  Addhuri Udupi Bhojana
4  https://www.zomato.com/bangalore/grand-village…           Grand Village

  online_order book_table  rate  votes      location      rest_type  \
0          Yes        Yes   4.1    775  Banashankari  Casual Dining
1          Yes         No   4.1    787  Banashankari  Casual Dining
2          Yes         No   3.8    918  Banashankari         Others
3           No         No   3.7     88  Banashankari    Quick Bites
4           No         No   3.8    166  Basavanagudi  Casual Dining

   approx_cost(for two people) listed_in(type)
0                        800.0          Buffet
1                        800.0          Buffet
2                        800.0          Buffet
3                        300.0          Buffet
4                        600.0          Buffet
```

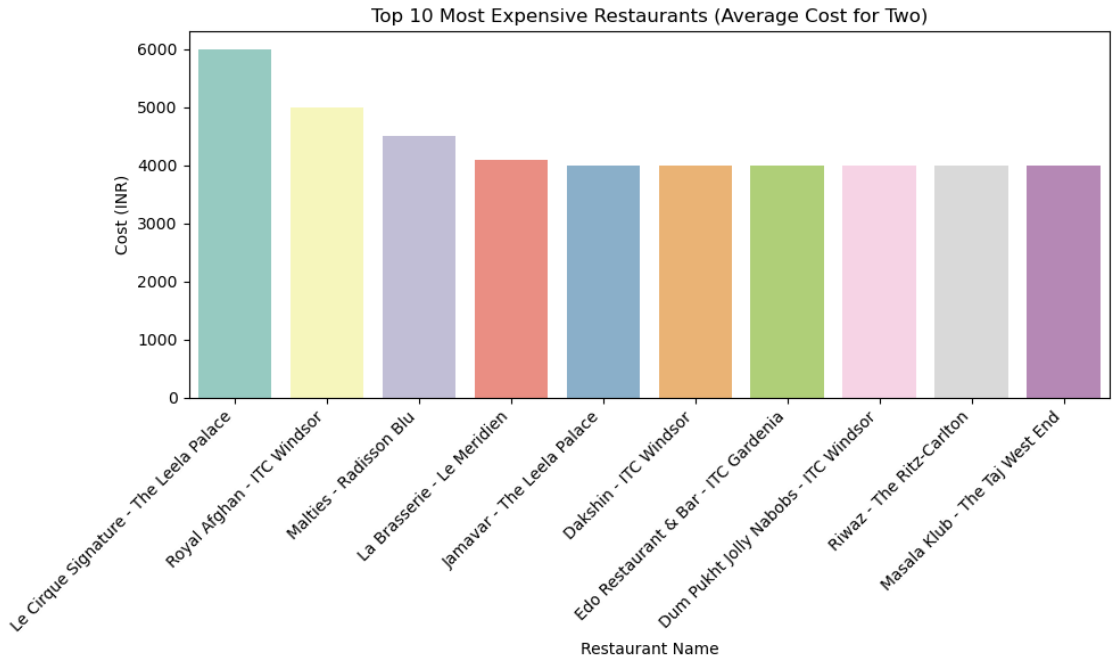
0.1.3 Top 10 Resturant based o highest avg cost

[458]:
```
import pandas as pd
import numpy as np

# Clean the 'approx_cost(for two people)' column if not done already
df['approx_cost(for two people)'] = (
    df['approx_cost(for two people)']
    .astype(str)
    .str.replace(',', '')
    .replace('nan', np.nan)
    .astype(float)
)

# Group by restaurant name and calculate average cost
cost_df = df.groupby('name').agg({
    'approx_cost(for two people)': 'mean'
}).dropna()

# Sort by cost descending and take top 10
top10_cost = cost_df.sort_values(by='approx_cost(for two people)',␣
 ↪ascending=False).head(10)
top10_cost = top10_cost.round(2)
top10_cost = top10_cost.rename(columns={'approx_cost(for two people)': 'Average␣
 ↪Cost for Two'})
```

```
# Optional: Clean non-ASCII characters
top10_cost = top10_cost.reset_index()
top10_cost['name'] = top10_cost['name'].str.encode('ascii', errors='ignore').
 ↪str.decode('ascii')

# Display the result
print(top10_cost)
```

```
                                    name  Average Cost for Two
0  Le Cirque Signature - The Leela Palace                6000.0
1              Royal Afghan - ITC Windsor                5000.0
2                  Malties - Radisson Blu                4500.0
3              La Brasserie - Le Meridien                4100.0
4              Jamavar - The Leela Palace                4000.0
5                   Dakshin - ITC Windsor                4000.0
6     Edo Restaurant & Bar - ITC Gardenia                4000.0
7    Dum Pukht Jolly Nabobs - ITC Windsor                4000.0
8                Riwaz - The Ritz-Carlton                4000.0
9          Masala Klub - The Taj West End                4000.0
```

[459]:
```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(10, 6))
sns.barplot(
    data=top10_cost,
    x='name',
    y='Average Cost for Two',
    hue='name',
    palette='Set3',
    legend=False
)

plt.title('Top 10 Most Expensive Restaurants (Average Cost for Two)')
plt.xlabel('Restaurant Name')
plt.ylabel('Cost (INR)')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```



[461]:
```
df.head()
```

[461]:

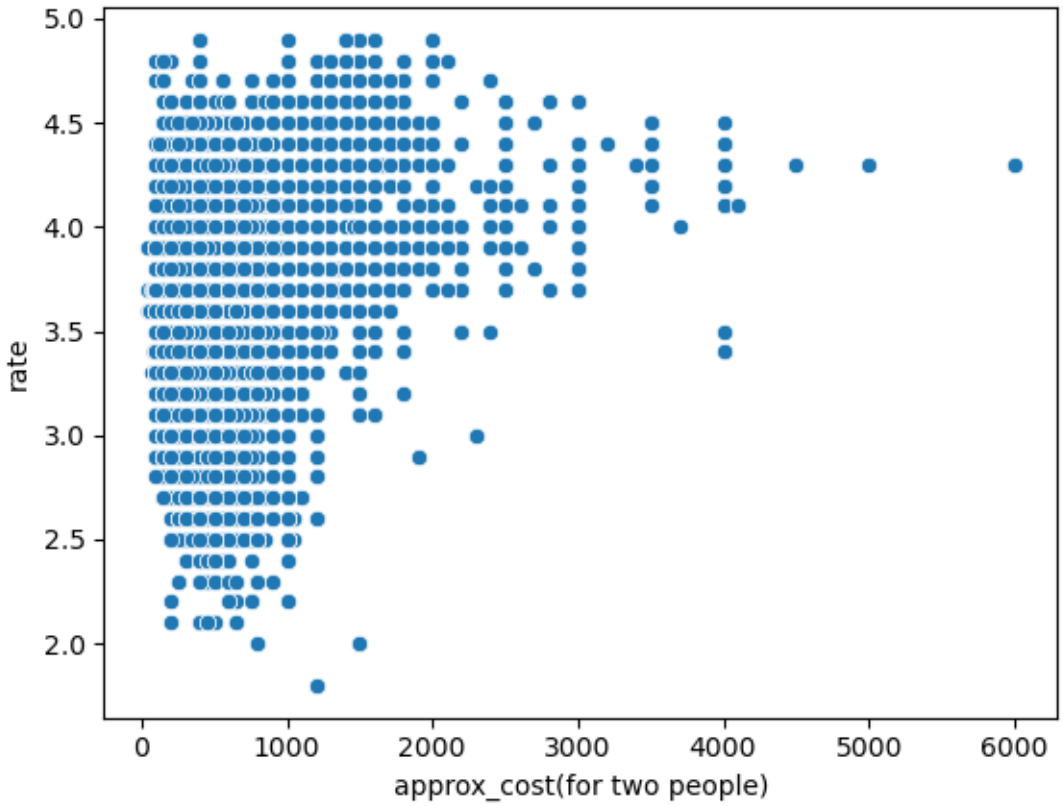
	url	name	online_order	book_table	rate	votes	location	rest_type	approx_cost(for two people)	listed_in(type)
0	https://www.zomato.com/bangalore/jalsa-banasha…	Jalsa	Yes	Yes	4.1	775	Banashankari	Casual Dining	800.0	Buffet
1	https://www.zomato.com/bangalore/spice-elephan…	Spice Elephant	Yes	No	4.1	787	Banashankari	Casual Dining	800.0	Buffet
2	https://www.zomato.com/SanchurroBangalore?cont…	San Churro Cafe	Yes	No	3.8	918	Banashankari	Others	800.0	Buffet
3	https://www.zomato.com/bangalore/addhuri-udupi…	Addhuri Udupi Bhojana	No	No	3.7	88	Banashankari	Quick Bites	300.0	Buffet
4	https://www.zomato.com/bangalore/grand-village…	Grand Village	No	No	3.8	166	Basavanagudi	Casual Dining	600.0	Buffet

0.1.4 1. Rating vs Cost

- Does expensive food get better ratings?

- Use scatter plot + correlation analysis.

[464]:
```
sns.scatterplot(data=df, x='approx_cost(for two people)', y='rate')
plt.show()
```



0.1.5 . Top Locations

- Which locations have the highest average ratings (top 10)?

[466]:
```
df.groupby('location')['rate'].mean().sort_values(ascending=False).head(10)
```

[466]:
```
location
Lavelle Road              4.106310
Koramangala 5th Block     3.983918
Church Street             3.980316
Koramangala 4th Block     3.880826
Residency Road            3.847478
Koramangala 7th Block     3.837245
MG Road                   3.833333
Indiranagar               3.817659
Richmond Road             3.786816
Koramangala 6th Block     3.776882
```

```
Name: rate, dtype: float64
```

[ ]:

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